

FlyByWire A32NX / A380X: upstream PR branches of the TCdr fork

Overview of the 24 features, their PR branches, dependencies and the rules they follow (2026-09-27)

Branch	24 branches pr/01 to pr/24 on the fork; develop 99f00b531 (2024.3.0-SNAPSHOT); master = upstream master (tag v2024.2.0)
Aircraft	A32NX and A380X (MSFS 2024)
Commits	fork https://github.com/TCdr/aircraft
Base	upstream flybywiresim/aircraft master (2baa2b35e) for every pr/* branch
Status	No PR opened yet. Every pr/* branch passes the CI checks (ESLint, unit tests, type-checked builds). PR 18 (and 19 on it) wait for the voice model licence check; PRs 22, 23 and 24 for their in-sim test.

Features, PR branches and documents

#	PR branch	Feature	Aircraft	Stacked on	Document
01	pr/01-a32nx-nd-weather-radar	A32NX native weather radar on the ND	A32NX	upstream master	01-a32nx-weather-radar.pdf
02	pr/02-a380x-nd-weather-radar	A380X weather radar on the ND and weather on the VD	A380X	PR 1	02-a380x-weather-radar-nd.pdf
03	pr/03-nd-wxr-mode-label	Weather radar mode on the ND	A32NX and A380X	PR 4	03-wxr-mode-label.pdf
04	pr/04-nd-terrain	Native TERR ON ND and the A380X VD terrain profile	A32NX and A380X	PR 2	04-terrain-on-nd.pdf
05	pr/05-a380x-surv-panel	A380X SURV controls, STATUS & SWITCHING and pedestal SURV panel	A380X	upstream master	05-a380x-surv-panel.pdf
06	pr/06-a380x-checklists-surv	A380X ECAM: WXR & TAWS and XPDR & TCAS lines sensed	A380X	PR 5	06-a380x-ecam-surv-lines.pdf
07	pr/07-tcas-ta-only-intruder-level	TCAS: RA-level intruders as TAs while RAs are inhibited	A32NX and A380X	upstream master	07-tcas-ta-only.pdf
08	pr/08-a32nx-mcdu-independent	Independent CPT and F/O MCDUs	A32NX	upstream master	08-a32nx-independent-mcdus.pdf
09	pr/09-nd-manuals-pass	Radar, terrain and VD after the A380 / A320 manuals	A32NX and A380X	PR 3	09-nd-nd-manuals-pass.pdf
10	pr/10-a380x-mfd-fcom-pages	A380X MFD FMS and SURV pages to the FCOM	A380X	upstream master	10-a380x-mfd-fcom-pages.pdf
11	pr/11-a380-takeoff-calculator	A380 takeoff calculator and company T.O data	A380X (flypad shared with the A32NX, see document 12)	PR 10	11-a380-takeoff-calculator.pdf
12	pr/12-a320-takeoff-calculator	A320 takeoff calculator and MCDU uplink takeoff data	A32NX (and the shared flypad screen of both aircraft)	PR 11	12-a320-takeoff-calculator.pdf
13	pr/13-pilot-stored-elements-persistence	Pilot stored elements kept across sessions	A380X (setting shared)	PR 10	13-pilot-stored-elements-persistence.pdf

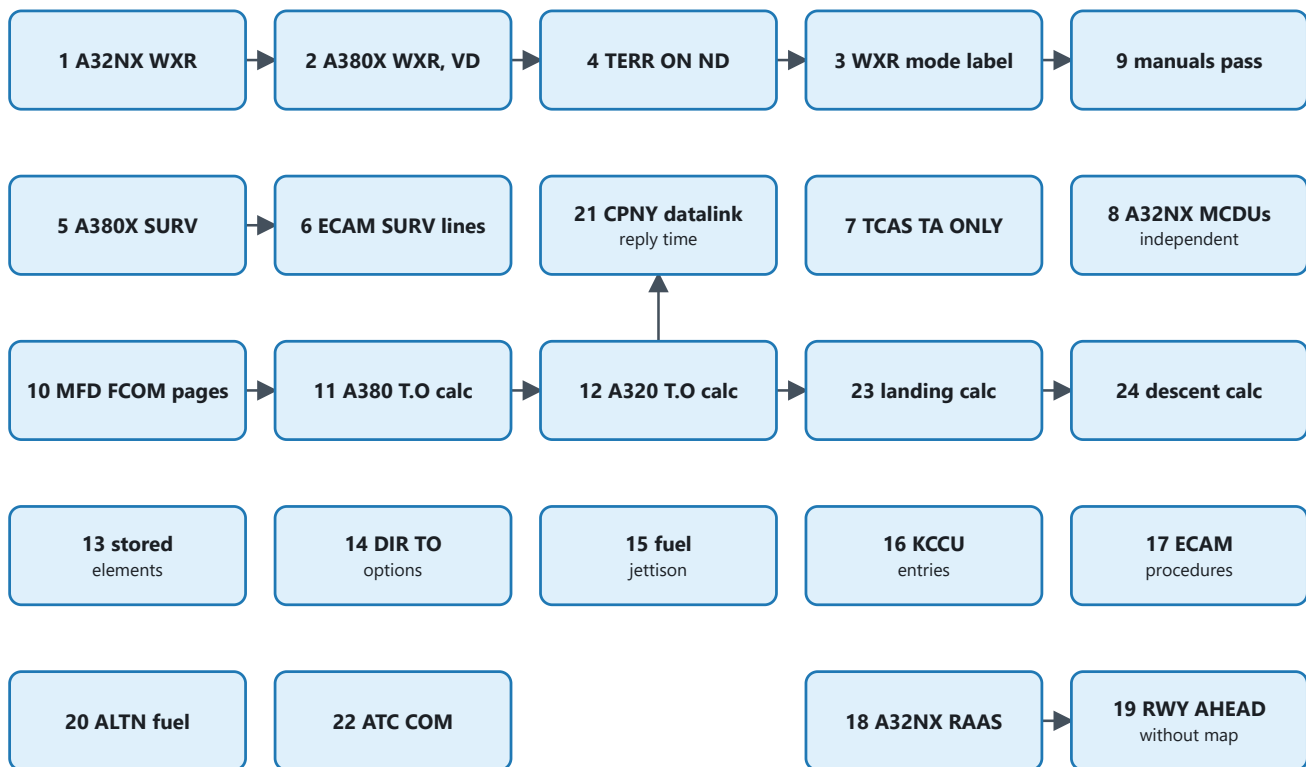
14	pr/14-direct-to-options	DIRECT TO: ABEAM points, CRS IN and CRS OUT	A380X (flight plan functions shared with the A32NX FMGC)	PR 10	14-direct-to-options.pdf
15	pr/15-fuel-jettison	Fuel jettison	A380X	PR 10	15-fuel-jettison.pdf
16	pr/16-kccu-keyboard-entry	KCCU keyboard entries	A380X (MFD, OIT, ND OANS panel)	PR 10	16-kccu-keyboard-entry.pdf
17	pr/17-ecam-procedures	ECAM procedures from the A380 FCOM	A380X	PR 10	17-ecam-procedures.pdf
18	pr/18-a32nx-raas	A32NX RAAS runway awareness callouts	A32NX (A380X OIT flyPad: settings option only)	upstream master	18-a32nx-raas.pdf
19	pr/19-a380x-oans-rwy-ahead-without-map	A380X OANS RWY AHEAD without an airport map	A380X	PR 18	19-a380x-oans-rwy-ahead-without-map.pdf
20	pr/20-a380x-altn-fuel-computation	A380X computed ALTN fuel and time	A380X (MFD FUEL&LOAD and ALTERNATE pages)	PR 10	20-a380x-altn-fuel-computation.pdf
21	pr/21-a380x-company-datalink-delay	A380X company datalink reply time	A380X (MFD, flyPad); A32NX MCDU unchanged	PR 12	21-a380x-company-datalink-delay.pdf
22	pr/22-a380x-mfd-atccom	A380X MFD ATC COM pages and the SD mailbox	A380X (MFD, SD)	PR 10	22-a380x-mfd-atccom.pdf
23	pr/23-efb-landing-calculator	flyPad landing calculator (A380 new, A320 reworked)	A380X and A32NX (flyPad, A380 FMC, A320 MCDU)	PR 12	23-efb-landing-calculator.pdf
24	pr/24-efb-descent-calculator	flyPad descent calculator	A380X and A32NX (flyPad, A380 FMC, A320 MCDU)	PR 23	24-efb-descent-calculator.pdf

Documents 01 to 17 were written from the feature branches (named in their header) before the clean pr/* branches were made; the pr/* branches hold the same features without the pr-docs files, with the develop fixes folded in.

Branch dependencies

A stacked branch contains the commits of the branch below it: open its PR once the one below is merged. The others are independent. All branches start from the upstream master 2baa2b35e.

■ Component added or changed by the feature
 ■ MSFS / simulator
 ■ Aircraft system (systems host, FMS, FWS)
 ■ Display / page
■ Data, LVar, storage
 ■ Crew, flypad, external program



Stacks (arrow: the next branch is built on it). 13, 14, 15, 16, 17, 20 and 22 are each stacked on 10; 1, 5, 7, 8, 10 and 18 start from the upstream master.

Working rules followed by every feature

- Manuals first: every behaviour follows the A380 / A320 FCOM or FCTM (or the vendor document) and the code comment names the page. Texts in `references/manuals`; A380 FCOM KAL 2011 (7156 pages), A320 FCOM 2019, A320 and A380 FCTM, Airbus A380 AC.
- No invented display behaviour; where MSFS cannot supply data, the limit is documented rather than faked.
- One branch per feature (feature/, fix/, improvement/) from develop; one clean pr/* branch per upstream PR; Conventional Commit titles; co-author trailer.
- Never in a commit: the `.env` files, `FwsAutoCallouts.ts`, `.idea/`, the `large-files` submodule.
- Checks of the upstream CI: ESLint (with Prettier), Rust fmt and clippy, the unit tests (vitest), the type-checked builds (FBW_TYPECHECK) of both aircraft; MFD / flyPad pages compared in a local browser harness; deployment into the installed packages with the layout.json kept consistent.

Release and open items

- master: the FBW upstream master 2baa2b35e itself (reset 2026-09-27), tagged v2024.2.0; it holds none of our features.
- develop: 2024.3.0-SNAPSHOT; holds features 1 to 24, the pedestal printer and the CI fixes, on top of the upstream master.
- Features 22 (ATC COM), 23 (landing calculator) and 24 (descent calculator) are merged into develop; the pedestal printer has no pr/* branch yet.
- Before opening: the aircraft-large-files PR for the MCDU screens (PR 8); the licence of the Piper voice model (PR 18); the in-sim test of PRs 22, 23 and 24.

A32NX native weather radar on the ND

A native WASM gauge (ndwxr) drawing MSFS weather radar data on both NDs

Branch	feature/a32nx-nd-weather-radar
Aircraft	A32NX
Commits	5 commits, tip 5b8a3604d (PR 1)
Base	upstream master 2baa2b35e
Status	In develop. Flown by the user. PR draft 01.

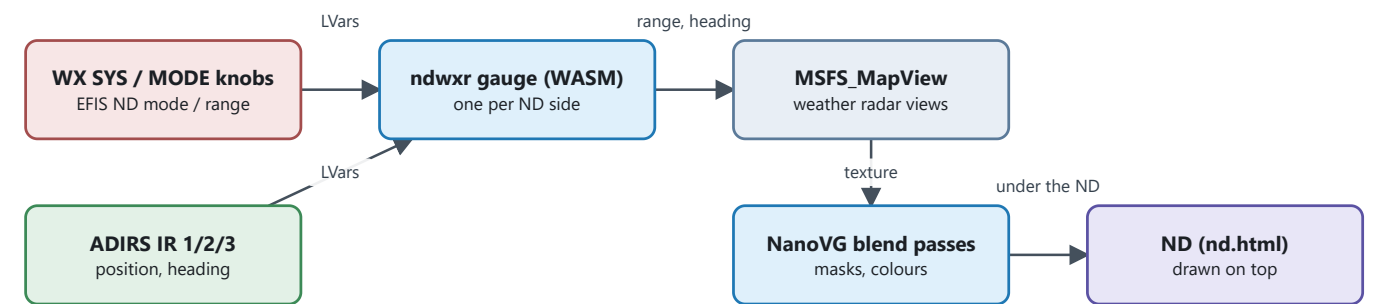
Purpose

The A32NX ND had no weather radar: the WX SYS / MODE knobs did nothing. The JS/Bing map radar path was found inconsistent between sessions; MSFS's native `MSFS_MapView` radar data was consistent, so the radar is a native gauge stacked on the ND.

Specification

- Drawn on the map pages of both NDs (ARC forward half disk, ROSE pages; the manuals pass extends it to ROSE ILS / VOR), aligned with the range rings; nothing on PLAN or with the radar off.
- WX SYS knob (SYS 1 / SYS 2 / OFF) and MODE knob: WX precipitation, WX+T precipitation + turbulence, TURB turbulence only, MAP a ground map (manuals pass).
- Colours by rain rate: green 0.7, yellow 4, red 12 mm/h; magenta turbulence proxy (rain rate > 20 mm/h, within 40 NM). Each threshold is a binary mask in a colour channel, blurred, saturated and squared: solid areas with crisp edges.
- Two independent NDs: CPT on AC ESS, F/O on AC 2; position from IR 1 / IR 2 or IR 3 per the ATT HDG knob.
- Build: igniter task `systems-ndwxr`, `wasm-opt` sign-extension lowering (as `terronnd`); `panel.cfg` gauges on VCockpit02 / VCockpit15.

■ Component added or changed by the feature ■ MSFS / simulator ■ Aircraft system (systems host, FMS, FWS) ■ Display / page
■ Data, LVar, storage ■ Crew, flypad, external program



Architecture

Limits of MSFS that shaped it

- No readback of the MapView texture (everything is blend passes); no turbulence data (proxy); no tilt or gain; at most 8 MapViews per module.

PR description (from the PR draft)

From `pr-drafts/01-wx-radar-a32nx.md`, section "Summary of Changes" onwards.

Summary of Changes

Adds the weather radar to the A32NX ND. It is drawn by a new native WASM gauge (`ndwxr`), stacked on the ND gauge of each side, from MSFS's own weather radar data (`MSFS_MapView`), so the ND JS code does not need the Bing map path.

- **Display:** ARC (forward half disk) and ROSE NAV (forward half), aligned with the ND's range rings, on the CPT and the F/O ND. Nothing is drawn on the other ND modes, or when the radar is off.
- **Controls:** the WX SYS knob (SYS 1 / SYS 2 / OFF) and the MODE knob: WX draws the precipitation, WX+T the precipitation and the turbulence, TURB only the turbulence, MAP nothing.
- **Colours:** airborne radar levels of rain rate: green 0.7, yellow 4, red 12 mm/h, magenta for turbulence. The engine's rain rate is noisy texel by texel, so each threshold is a binary mask (in a colour channel of the MapView) that is blurred, saturated and squared, which gives solid areas with crisp edges instead of grain.
- **Turbulence** is a proxy: MSFS gives no turbulence data, only rain rate. Rain rate above 20 mm/h is drawn magenta, out to 40 NM.
- **Two independent NDs:** the CPT ND is on the AC ESS bus, the F/O ND on AC 2. The position source follows the ADIRS rule of the ND: IR 1 for the CPT, IR 2 for the F/O, or IR 3 when the ATT HDG knob routes it to that side. ND mode and range come from `A32NX_EFIS_{L,R}_ND_MODE` / `_RANGE`. The two WX knobs are shared.
- **Ground:** like a real weather radar it does not transmit on the ground. Nothing is drawn while both main gear legs are compressed on either LGCIU. The engine already stabilises the beam in pitch and bank.
- **Build:** new igniter task `systems-ndwxr` (`build-a32nx:ndwxr`), the module is built with the same `wasm-opt` sign-extension lowering as `terrond`, `panel.cfg` gets the gauge on the two ND blocks (`VCockpit02` / `VCockpit15`, with the side as the last parameter), and `docs/README.md` lists the new folder.

Limits of MSFS that shaped it: no readback of the MapView texture (so everything is blend passes), no turbulence data, no tilt or gain, and the radar image is only drawn when it is airborne with weather around (use the MSFS weather presets to test).

Cockpit API Changes

No new LVars, no LVar changed. The gauge reads: `L:A32NX_EFIS_{L,R}_ND_MODE`, `L:A32NX_EFIS_{L,R}_ND_RANGE`, `L:XMLVAR_A320_WeatherRadar_Sys`, `L:XMLVAR_A320_WeatherRadar_Mode`, `L:A32NX_ADIRS_IR_{1,2,3}_LATITUDE` / `_LONGITUDE` / `_TRUE_HEADING`, `L:A32NX_ATT_HDG_SWITCHING_KNOB`, `L:A32NX_ELEC_AC_ESS_BUS_IS_POWERED`, `L:A32NX_ELEC_AC_2_BUS_IS_POWERED`, `L:A32NX_LGCIU_{1,2}_{LEFT,RIGHT}_GEAR_COMPRESSED`.

Screenshots (if necessary)

TO ADD (same camera, zoom and time of day, before/after): the CPT ND in ARC and in ROSE NAV with rain and a thunderstorm at 40 and 80 NM; WX and WX+T side by side; the F/O ND showing a different range; the ND on the ground (no radar). Before: the same ND with nothing drawn.

References

- MSFS SDK, `MSFS_MapView.h` native weather radar API (implementation, not real-life behaviour).
- **TO ADD** (the guide asks for real-life references): the A320 FCOM/PSCOM pages showing the weather radar on the ND (colour coding of precipitation and turbulence, WX / WX+T / TURB modes, WXR not transmitting on the ground), with screenshots.

Additional context

The native MapView weather data was found to be far more consistent across sessions than the JS/Bing map path, which is why the radar is a WASM gauge. `ndwxr` serves both ND gauges from one module (state per gauge context),

and each ND creates two MapViews (precipitation and turbulence). MSFS allows at most 8 MapViews per module, which matters for the TERR ON ND PR that reuses this module.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. Start a flight in the air with weather (MSFS weather presets: rain, thunderstorm). Set the ND to ARC or ROSE NAV, WX SYS on SYS 1 and MODE on WX: precipitation appears in green/yellow/red, aligned with the range rings. Change the range (10 to 320) and switch ARC/ROSE NAV: the image stays aligned.
2. MODE on WX+T: magenta turbulence cells appear over strong cells (within 40 NM). TURB: only the magenta. MAP: nothing.
3. WX SYS on OFF, or the ND on PLAN / ROSE ILS / ROSE VOR: no radar.
4. On the ground (gear compressed): no radar. After take-off it comes back.
5. F/O ND: set another range and mode than the CPT ND: both show their own image. Switch off AC ESS (CPT) and AC 2 (F/O): the corresponding radar goes off. Try the ATT HDG knob (CAPT/F/O 3) on the position source.
6. Restart MSFS with the same weather and check the same cells appear (consistency between sessions).
7. Check the ND lines, range rings, traffic and flight plan stay visible over the weather.

A380X weather radar on the ND and weather on the VD

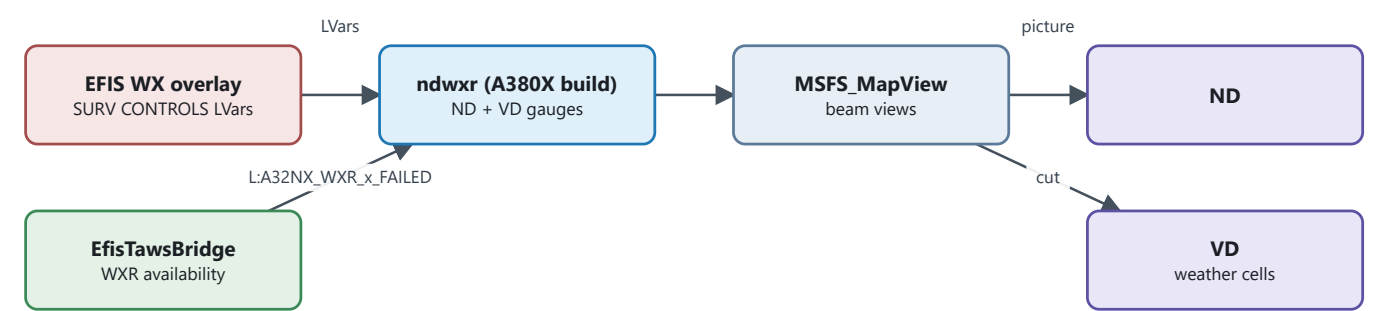
The ndwxr gauge for the A380X, following the EFIS WX overlay and the SURV CONTROLS

Branch	feature/a380x-nd-weather-radar
Aircraft	A380X
Commits	tip 416a0f316 (PR 2), on PR 1
Base	feature/a32nx-nd-weather-radar
Status	In develop. Flown. PR draft 02.

Specification

- Availability (EfisTawsBridge): both WXR systems were hard-coded failed (WXR INOP). Each is now failed only with its AESU reset or its power lost (AC ESS system 1, AC 4 system 2).
- ND radar on both A380X NDs, shown with this side's WX overlay and a WXR & TAWS system selected and not failed; range index 0 (OANS) hides it; position IR 1 / 2 or IR 3 per ATT HDG.
- Follows the SURV CONTROLS: WXR, TURB (WX / WX+T; no TURB-only mode on the A380), MODE MAP, WX ON VD.
- VD weather (WX ON VD): the vertical MapView mode is one thin beam, so the VD weather is stylised from the ND radar views along the cut (green / yellow columns, red core, no magenta: FCOM DSC-31-20-40-10).
- 30 s buffer fill after the radar starts transmitting (FCOM operational recommendations).

Component added or changed by the feature MSFS / simulator Aircraft system (systems host, FMS, FWS) Display / page
Data, LVar, storage Crew, flypad, external program



Architecture

The A380's AUTO-mode off-path display was built from a third MapView in the engine's top-view mode and removed: the engine runs one radar per aircraft and that view corrupted the beam views. Both aircraft show the same picture.

PR description (from the PR draft)

From `pr-drafts/02-wx-radar-a380x-vd.md`, section "Summary of Changes" onwards.

Summary of Changes

Adds the weather radar to the A380X NDs with the native `ndwxr` gauge of the A32NX PR, and the weather to the VD.

- **Availability (EfisTawsBridge):** both WXR systems were hard-coded as failed, so the ND always showed WXR INOP as soon as the WX overlay was selected. Each WXR is now failed only while its AESU is reset or its power supply is lost (AC ESS for system 1, AC 4 for system 2), like GPWS.
- **ND radar:** built for the A380X (`build.sh --a380x, ndwxr_A380X.wasm, build-a380x:ndwxr`, igniter task) and added to both ND screens (`Vcockpit07` CPT, `Vcockpit08` F/O, 768x1024). It follows the A380X's rules: shown when this side's EFIS panel has the WX overlay selected and the WXR & TAWS system selected on the SURV panel is not failed; the ND range table is `[-1, 10 .. 640]` and range index 0 (the OANS map) hides the radar; the ND is on with the DC buses of its display unit (`CdsDisplayUnit` mapping); IR 1/2 feed the CPT/F/O with the ATT HDG knob in NORM, IR 3 otherwise. The ND pages are shared with the A32NX, so the geometry is the same.
- **SURV CONTROLS:** the radar follows the WXR, TURB, MODE (MAP) and WX ON VD buttons (`L:A380X_WXR_OFF`, `_TURB_OFF`, `_MODE_MAP`, `_VD_OFF`, written by the SURV PR). There is no TURB-only mode on the A380X: TURB off gives WX, on gives WX+T. (`PRED WS` and `GAIN` are written by the SURV page too but nothing reads them: MSFS's radar has no predictive windshear or gain.)
- **VD weather (WX ON VD):** the vertical MapView mode of MSFS is a single thin beam (and tilting it crashes the gauge), so the VD weather is stylised from the ND's own radar views: the texel columns near the heading line are rotated onto the range axis and drawn as cells (green and yellow columns with rounded tops, a red core; no magenta: the real VD does not display turbulence, FCOM DSC-31-20-40-10). The heights are fractions of the VD altitude span above the aircraft, placed on the ADR baro-corrected altitude like the VD's own symbol. Single-texel specks and radar sweep streaks are filtered.
- **Mouse on the ND:** the `ndwxr` gauges are stacked over the ND in the same Vcockpit page and only draw; the ND stylesheet makes them transparent to the mouse (`pointer-events: none`), otherwise they catch every click on the OANS moving airport map (map drag, BTV runway and exit selection). Commit `248cb26bd`.

Cockpit API Changes

Reads (written by the SURV PR, which documents them): `L:A380X_WXR_OFF`, `L:A380X_WXR_TURB_OFF`, `L:A380X_WXR_MODE_MAP`, `L:A380X_WXR_VD_OFF` (Bool, one per SURV CONTROLS button). Also reads `L:A380X_EFIS_{L,R}_ACTIVE_OVERLAY`, `L:A32NX_WXR_TAWS_SYS_SELECTED`, `L:A32NX_WXR_{1,2}_FAILED`, `L:A32NX_VD_{1,2}_RANGE_LOWER/UPPER` and the ND mode/range LVars. No other LVar added or changed. `panel.cfg` (A380X cockpit part) gets the gauges on the two ND blocks.

Screenshots (if necessary)

TO ADD: before/after of the A380X ND with the WX overlay selected (before: WXR INOP, no weather); CPT with weather and F/O with another range/overlay; the VD with WX ON VD; the SURV CONTROLS page next to the resulting ND.

References

- MSFS SDK, `MSFS_MapView.h` native weather radar API.
- **TO ADD:** A380 FCOM/documentation for the WX overlay of the ND, the WXR & TAWS system selection and the weather on the VD, with screenshots.

Additional context

The VD weather is an approximation (MSFS has no vertical weather data): the shapes are made from the horizontal radar image, they are not real cell heights. Say so in the review if you would rather not ship that part, the ND radar

works without it.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. In flight with weather, select the WX overlay on the CPT EFIS panel and a WXR & TAWS system on the SURV panel: weather appears on the CPT ND (ARC and ROSE NAV, ranges 10 to 640), no WXR INOP message. Do the same on the F/O side with another range.
2. SURV CONTROLS page: WXR off removes it, TURB toggles WX / WX+T, MODE MAP removes the precipitation.
3. Range index 0 (OANS) and the PLAN mode hide the radar.
4. WX ON VD: weather cells appear on the VD around the aircraft altitude, and disappear with WX ON VD off.
5. Reset the AESU of a WXR system, or remove AC ESS / AC 4: that system is failed (WXR INOP) and its weather goes.
6. ATT HDG knob away from NORM: the position source changes, no glitch.

Weather radar mode on the ND

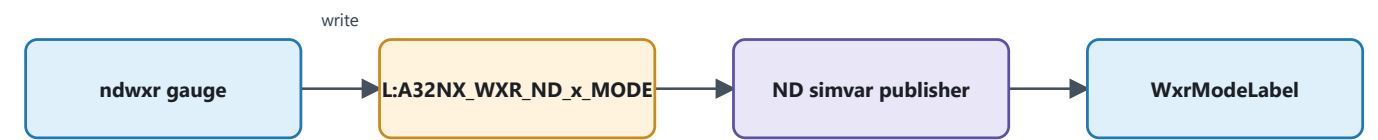
WX, WX+T, TURB, MAP or WXR OFF shown on the map pages

Branch	feature/nd-wxr-mode-label
Aircraft	A32NX and A380X
Commits	a5ecd08cf (PR 3)
Base	feature/nd-terrain
Status	In develop. PR draft 03.

Specification

- The gauge publishes the mode as `L:A32NX_WXR_ND_{L,R}_MODE` (0 none, 1 WX, 2 WX+T, 3 TURB, 4 MAP, 5 WXR OFF), only while the radar is selected on a map page and not replaced by the terrain.
- The ND draws it with the new `WxrModeLabel1` (green, right of the page; WXR OFF white on the A32NX, A320 FCOM DSC-34-SURV-30-30). The A380X shows its own list: WX / MAP / WXR OFF (A380 FCOM DSC-34-20-30-20).

■ Component added or changed by the feature ■ MSFS / simulator ■ Aircraft system (systems host, FMS, FWS) ■ Display / page
■ Data, LVar, storage ■ Crew, flypad, external program



Architecture

PR description (from the PR draft)

From `pr-drafts/03-wx-mode-label.md`, section "Summary of Changes" onwards.

Summary of Changes

Nothing on the ND told the crew which radar mode was selected. The mode (WX, WX+T, TURB or MAP) is now shown in green on the right of the ARC and ROSE NAV pages while the radar is on, on both aircraft.

- The native radar gauge already knows whether the radar is selected and shown (the WX SYS knob on the A32NX; the WX overlay, the WXR & TAWS system and the SURV CONTROLS buttons on the A380X), so it publishes the text as a number in `A32NX_WXR_ND_{L,R}_MODE` (0 none, 1 WX, 2 WX+T, 3 TURB, 4 MAP), only while the radar is selected on an ARC or ROSE NAV page and not while the terrain takes its place.
- The ND reads it through its simvar publisher and the new `WxrModeLabel` component draws the text, so the gating rules are not duplicated in the ND code.
- The A380X has no TURB-only mode: its SURV CONTROLS page gives WX, WX+T and MAP.
- The text is also shown on the ground, where the radar itself does not draw.

Cockpit API Changes

New LVars, written by the `ndwxr` gauge and read by the ND. Already documented in `fbw-a32nx/docs/a320-simvars.md` (new "Weather Radar (ATA 34)" section) and `fbw-a380x/docs/a380-simvars.md` (new "Surveillance ATA 34" section), in the commit of this PR:

Var	Type	Description
<code>L:A32NX_WXR_ND_L_MODE</code>	Enum	Weather radar mode text of the left (CPT) ND: 0 none, 1 WX, 2 WX+T, 3 TURB, 4 MAP
<code>L:A32NX_WXR_ND_R_MODE</code>	Enum	Same for the right (F/O) ND

Screenshots (if necessary)

TO ADD: ARC and ROSE NAV with the label in each mode (WX, WX+T, TURB, MAP), and without the radar (no label). Before: same page, no label.

References

TO ADD: real ND photo/FCOM screenshot showing the weather radar mode text (position and colour) on the A320 and the A380 ND.

Additional context

Position: x 744, y 590 of the ND svg, `Green FontSmall1`, right-aligned, after `TerrainMapThresholds`. The labels are not localised (like the other ND texts).

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. A32NX: WX SYS on SYS 1, ND in ARC then ROSE NAV: the mode text follows the MODE knob (WX, WX+T, TURB, MAP). WX SYS OFF: no text.
2. Other ND modes (PLAN, ROSE ILS, ROSE VOR): no text.
3. Turn TERR ON ND on (with the terrain PR): the text goes while the terrain is drawn.
4. A380X: WX overlay + WXR & TAWS system selected: text follows the SURV CONTROLS TURB / MODE buttons (WX, WX+T, MAP). Range index 0 (OANS): none.
5. On the ground the text is shown although the radar does not draw.

Native TERR ON ND and the A380X VD terrain profile

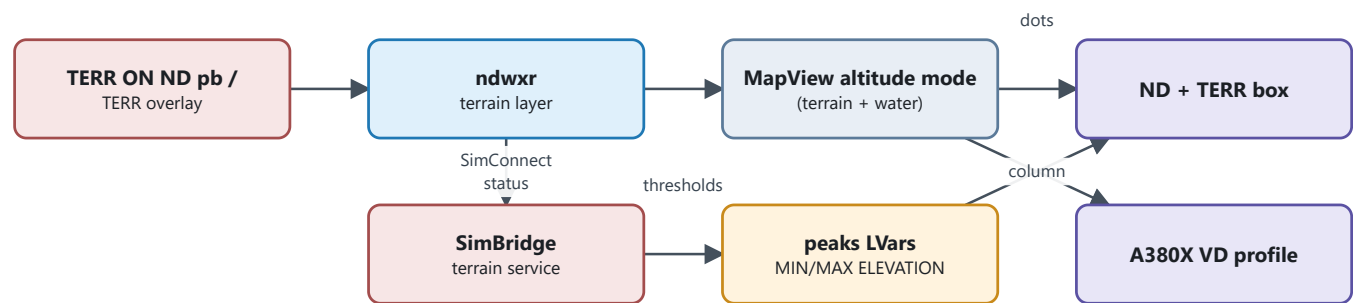
EGPWS-style terrain from MSFS terrain data, without SimBridge rendering

Branch	feature/nd-terrain
Aircraft	A32NX and A380X
Commits	tip acffe345b (PR 4)
Base	feature/a380x-nd-weather-radar
Status	In develop. Flown. PR draft 04.

Specification

- An altitude-mode MapView colours the terrain by height relative to the aircraft in 250 ft bands (-29 500 to +2 500 ft, 128 entries); the EGPWS look is built from the bands: red from 2000 ft above, yellow from 1000 ft above to 500 ft below (250 ft gear down), green to 1000 / 2000 ft below; random stipple dots from blend-only passes.
- Reference altitude 30 s ahead when descending faster than 1000 ft/min (A320 FCOM DSC-31-45).
- Water from a second altitude view, drawn as teal dots. Map rotated by the ND true heading; every map page but PLAN, also on the ground, in place of the weather.
- A32NX gated by the TERR ON ND pb; A380X by the TERR overlay with a WXR & TAWS system selected.
- A380X VD terrain profile: a third gauge per ND side, the terrain column along the cut stretched over the plot and filled brown.
- Peaks box: figures from SimBridge over SimConnect (ndwxr replaces the stock terronnd as SimBridge's client); terronnd no longer instantiated.

■ Component added or changed by the feature ■ MSFS / simulator ■ Aircraft system (systems host, FMS, FWS) ■ Display / page
■ Data, LVar, storage ■ Crew, flypad, external program



Architecture

MSFS limits found in-sim: at most 8 MapViews per module (a ninth crashes the draw), no growing the module memory while drawing (16 MB reserved).

PR description (from the PR draft)

From `pr-drafts/04-terrain-on-nd.md`, section "Summary of Changes" onwards.

Summary of Changes

Terrain on the ND (TERR ON ND) without SimBridge, drawn by the `ndwxr` module for both aircraft from MSFS's own terrain data, plus a terrain profile on the A380X VD.

How the ND terrain works

- An altitude-mode MapView colours the terrain by the aircraft's altitude minus the terrain height in 250 ft bands, over -29,500 to +2,500 ft (128 entries). The EGPWS look is built from the bands: dense red from 2000 ft above the aircraft, dense/medium yellow from 1000 ft above, medium yellow down to 500 ft below (250 ft with the gear down), dense green to 1000 ft below (500 ft), light green to 2000 ft below and nothing under that. The dots are a random stipple (~1 px cells, ~70% lit when dense) made with blend-only passes (ordered compare against a noise image, sharpened by repeated squaring), as MSFS gives a gauge no way to read the texture back.
- Water never takes the altitude list's colours (it always gets the first entry, like terrain far above), so a second altitude view with a two-entry list tells water from land; the water is wiped from the terrain and redrawn as cyan-blue dots at every altitude.
- The map is rotated by the ND's own true heading. ARC shows the forward half disk, the ROSE modes the whole disk; every map page but PLAN, also on the ground, in place of the weather.
- A32NX: gated by `A32NX_EGPWC_ND_{L,R}_TERRAIN_ACTIVE` (the TERR ON ND pb), gear from `A32NX_EGPWC_GEAR_IS_DOWN`. A380X: gated by the ND overlay being TERR with a WXR & TAWS system selected and not failed.
- Terrain far above the range's top does not clamp to the first entry in MSFS (it comes out as rings of the other entries), which is why the range reaches -29,500 ft and every band above +2000 ft is red: nothing on Earth is higher above the aircraft.

A380X VD terrain profile: a third `ndwxr` gauge per ND (`LV / RV` in `panel.cfg`, its own surface). The heading-line column of a terrain view whose altitude range follows the VD's limits is stretched over the plot height, compared with a vertical ramp and filled brown; water is cyan from sea level down. Needs the TERR SYS button of the SURV page not to be OFF.

MSFS limits that shaped the module (found by crashing the gauge draw in-sim): a module can have at most 8 MapViews (a ninth is created without complaint and crashes the draw; hiding views does not help). The A32NX uses 4 per ND; the A380X has 2 per ND, which are either the weather pair or the terrain pair (never wanted together) and are reconfigured when the crew switches, and 2 per VD terrain gauge. Growing the module's memory while it draws also crashes it, so `build.sh` reserves 16 MB up front.

Peaks box (PR 9's commit 670dd76b1, 2026-09-23): the TERR box with the highest and lowest elevation of the range needs real elevations, which a gauge cannot read back from MSFS, so its figures come from SimBridge, over SimConnect: SimBridge's terrain service only renders while a SimConnect client takes part (its HTTP `renderingThresholds` endpoint stays empty while it is connected to the sim; an HTTP poller was tried first and reverted), and the stock `terronnd` gauge was that client. `ndwxr` now takes its place from the first ND gauge installed: it creates and writes the aircraft status block `FBW_SIMBRIDGE_EGPWC_AIRCRAFT_STATUS` every 100 ms (the content of `terronnd's collection.cpp`, with the terrain reported on per side exactly when the module selects it), subscribes to the two threshold areas SimBridge writes per ND and publishes the figures on the four `L:A32NX_EGPWC_ND_{side}_TERRAIN_{MIN,MAX}_ELEVATION(_MODE)` LVars that `TerrainMapThresholds` already shows, while terrain is selected on that side on a map page and the figures are fresh (5 s) and for the ND's range and mode; -1 otherwise. It reconnects every 5 s and disconnects when the owning gauge is killed. `build.sh` adds the SimConnect SDK include path; the client data layouts are `terronnd's simbridge.h`. Without SimBridge the box stays empty and the terrain is drawn as before. (Peaks mode, the display when all terrain is far below the aircraft, uses the same figures: PR 9.)

terronnd retired (uncommitted, 2026-09-23): the stock `terronnd` gauge is removed from both `panel1.cfg` (the `was` is still built, nothing instantiates it). Reasons found in test with SimBridge 0.7.0 on MSFS 2024: its picture is not clipped at the compass arc (the old `ARC_CLIP` dome), it is never expired (the last frame stays after SimBridge closes), and on the A380X its SimConnect status reports terrain on for an ND whenever the VD wants it, which SimBridge falls back to after two minutes without HTTP posts (a stationary aircraft), so the F/O terrain stayed on with the overlay off. The native picture is the only one now; SimBridge only supplies the peaks figures. The A32NX gets `TawsStatusBridge` (systems host) to post the aircraft status over HTTP as the A380X does (terronnd used to send it over SimConnect; the SimConnect status block SimBridge takes the aircraft's position from is written by `ndwxr` again, see the peaks box), and both hosts re-post it every 30 s so SimBridge never drops the HTTP source. The ARC picture (terrain and weather) leaves the ND's bottom-right corner free (same corner as CanvasMap's `ARC_CLIP`) for the TERR box and the radar mode text.

Not done: the VD profile follows the heading line and not the flight plan path, and switching between weather and terrain on the A380X ND takes about a second.

Cockpit API Changes

No new LVars. The terrain gauge reads `L:A32NX_EGPWC_ND_{L,R}_TERRAIN_ACTIVE`, `L:A32NX_EGPWC_GEAR_IS_DOWN`, `L:A32NX_GPWS_TERR_OFF`, `L:A32NX_TERR_{1,2}_FAILED`, `L:A32NX_WXR_TAWS_SYS_SELECTED`, `L:A380X_EFIS_{L,R}_ACTIVE_OVERLAY` and `L:A32NX_VD_{1,2}_RANGE_LOWER/UPPER` (the last two pairs already exist), `panel1.cfg` gets the VD gauges.

Screenshots (if necessary)

TO ADD: TERR ON ND on both aircraft in ARC and ROSE, at the same place before/after (before: no terrain without SimBridge), e.g. KASE (Aspen, 7,820 ft) at 40 and 320 NM, and LSZC; sea and lakes in blue; gear up/down colour change; the A380X VD terrain profile; weather on one ND and terrain on the other.

References

- MSFS SDK, `MSFS_MapView.h` altitude view mode (implementation).
- TO ADD:** the Honeywell EGPWS / A320 & A380 FCOM references for the terrain colours and the altitude thresholds (the values above), and for the gear-down thresholds; if the blue water cannot be referenced, drop it (it was added because the reference aircraft in the sim show it).

Additional context

The stock `terronnd` module is only a client of the external SimBridge program; its source is not changed by this PR, but the gauge is no longer instantiated (removed from both `panel1.cfg`, see above), so there is one terrain layer. Say in the review if the `terronnd` build task should go too. After the range fix the A380X VD terrain profile still uses a range that ends at the top of its plot (same weakness as the ND had), it was not affected in tests at Aspen where the plot sits above the terrain.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

- A32NX: TERR ON ND pb ON, ND in ARC or ROSE NAV: terrain in red/yellow/green dots around the aircraft, rotating with the heading. Check at high terrain: on the ground at KASE (Aspen) at 40 and 320 NM there must be red.
- Lower the gear: the medium-yellow/green limits move (250/500 ft instead of 500/1000 ft below). Water is blue dots.
- PLAN: no terrain. On the ground the terrain is drawn.
- A380X: TERR overlay on the EFIS panel and a WXR & TAWS system selected: same on the ND; VD shows the terrain profile (TERR SYS not OFF); with TERR SYS OFF the VD says NO TERR DATA AVAILABLE (see the SURV PR).
- A380X: WX on the CPT ND and TERR on the F/O ND at once, then swap them (about 1 s to switch).
- Weather and terrain never both show on the same ND; the weather comes back when TERR is turned off.

A380X SURV controls, STATUS & SWITCHING and pedestal SURV panel

The MFD SURV pages and the pedestal buttons made functional

Branch	feature/a380x-surv-panel
Aircraft	A380X
Commits	3 commits, tip a1c0afd4b (PR 5)
Base	upstream master 2baa2b35e
Status	In develop. PR draft 05.

Specification

- SURV CONTROLS WXR block (WXR, PRED WS, TURB, GAIN, MODE, WX ON VD) writes LVars the radar reads; inhibited only when the selected WXR system is failed.
- STATUS & SWITCHING: WXR DISPLAY shows the real failures; XPDR/TCAS SYS 1 / 2 select
L:A32NX_TRANSPONDER_SYSTEM.
- Pedestal SURV panel: INDICATOR_POWERED was never defined, so no button worked; now DC 1. WXR TAWS SYS 1/2, XPDR TCAS SYS 1/2, G/S MODE, TCAS ABV / BLW / TA ONLY in step with the MFD pages.
- VD messages: NO TERR DATA / NO WX DATA / both, per case.

LVar	Meaning
L:A380X_WXR_OFF , _TURB_OFF , _MODE_MAP , _VD_OFF , _PRED_WS_OFF , _GAIN_MAN	SURV CONTROLS WXR buttons
L:A380X_TCAS_ALERT_LEVEL , _ALT_SELECT (+ _REQUEST)	TCAS settings mirrored for the pedestal

PR description (from the PR draft)

From `pr-drafts/05-a380x-surv.md`, section "Summary of Changes" onwards.

Summary of Changes

MFD SURV CONTROLS: the WXR block was a stub (every button greyed out, "FIXME replace with appropriate events"). Its buttons (WXR, PRED WS, TURB, GAIN, MODE, WX ON VD) now write LVars that the native weather radar reads, and are only inhibited when the selected WXR system is really failed.

STATUS & SWITCHING: the WXR DISPLAY items show the real failure of each system, and the XPDR/TCAS SYS 1 / SYS 2 buttons select the transponder system (`L:A32NX TRANSPONDER_SYSTEM`, the one the TCAS reads) and light from it.

Pedestal SURV panel: the buttons use the backlight indicator template, which runs the button code and lights the button only if `INDICATOR_POWERED` is defined, and `pedestal.xml` never defined it (the EFIS, FCU and KCCU files do), so none of them ever worked. It is now the DC 1 bus, like the backlight. The buttons are linked to the MFD pages: WXR TAWS SYS 1/2 toggle `L:A32NX_WXR_TAWS_SYS_SELECTED`, XPDR TCAS SYS 1/2 set the transponder system, G/S MODE toggles the GPWS G/S inhibit, each lit from the same LVar. TCAS ABV / BLW / TA ONLY follow and change the SURV CONTROLS TCAS settings: the TCAS computer mirrors the settings to LVars for the lights and applies the button requests to the same events the page publishes.

VD messages: TERR SYS OFF removed both terrain and weather from the message logic, it always showed "NO TERR AND WX DATA AVAILABLE". TERR SYS OFF now only gives NO TERR DATA AVAILABLE, a failed or OFF WXR or WX ON VD OFF only gives NO WX DATA AVAILABLE, and the combined message is kept for both.

Known limit: PRED WS and GAIN MAN are written but nothing reads them (the MSFS radar has no predictive windshear or gain).

Cockpit API Changes

New LVars, already documented in `fbw-a380x/docs/a380-simvars.md` (new "Surveillance ATA 34" section, in the commits of this PR):

Var	Type	Description
<code>L:A380X_WXR_OFF</code>	Bool	SURV CONTROLS WXR button: weather radar off
<code>L:A380X_WXR_TURB_OFF</code>	Bool	SURV CONTROLS TURB button: turbulence detection off
<code>L:A380X_WXR_MODE_MAP</code>	Bool	SURV CONTROLS MODE button: MAP mode
<code>L:A380X_WXR_PRED_WS_OFF</code>	Bool	SURV CONTROLS PRED WS button: predictive windshear off (not read by anything yet)
<code>L:A380X_WXR_GAIN_MAN</code>	Bool	SURV CONTROLS GAIN button: manual gain (not read by anything yet)
<code>L:A380X_WXR_VD_OFF</code>	Bool	SURV CONTROLS WX ON VD button: weather on the VD off
<code>L:A380X_TCAS_ALERT_LEVEL</code>	Number	Current TCAS alert level setting of the SURV CONTROLS page (mirrored for the pedestal lights)
<code>L:A380X_TCAS_ALT_SELECT</code>	Number	Current TCAS ABV/BLW altitude selection (mirrored for the pedestal lights)
<code>L:A380X_TCAS_ALERT_LEVEL_REQUEST</code>	Number	Pedestal button request: wanted alert level + 1, 0 = no request (cleared by the TCAS computer)
<code>L:A380X_TCAS_ALT_SELECT_REQUEST</code>	Number	Pedestal button request: wanted altitude selection + 1, 0 = no request

Enum meanings as documented: `L:A380X_TCAS_ALERT_LEVEL` 0 STBY / 1 TA ONLY / 2 TA/RA, `L:A380X_TCAS_ALT_SELECT` 0 NORM / 1 ABV / 2 BLW; the WXR LVars are true for OFF / MANUAL / MAP.

Screenshots (if necessary)

TO ADD: the SURV CONTROLS page (WXR block enabled) and STATUS & SWITCHING before/after; the pedestal SURV panel lit (before: all buttons dark); the VD messages for each case.

References

TO ADD: A380 FCOM/FCTM pages for the SURV panel and the SURV CONTROLS / STATUS & SWITCHING MFD pages (button functions, WXR & TAWS and XPDR & TCAS system selection, TCAS ABV/BLW/TA ONLY, VD no-data messages), with screenshots.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. MFD SURV CONTROLS: WXR, TURB, MODE, WX ON VD buttons change the weather on the ND/VD (with the weather PR); PRED WS / GAIN toggle without effect. With the selected WXR system failed (reset its AESU) the WXR buttons are inhibited.
2. MFD STATUS & SWITCHING: WXR DISPLAY shows the failures; XPDR/TCAS SYS 1 / SYS 2 select the transponder system and light.
3. Pedestal SURV panel: with DC 1 powered the buttons light; WXR TAWS SYS 1/2, XPDR TCAS SYS 1/2, G/S MODE, TCAS ABV / BLW / TA ONLY work and stay in step with the MFD pages (press on the pedestal, look at the MFD, and the other way round). Remove DC 1: lights off.
4. VD messages: TERR SYS OFF -> NO TERR DATA AVAILABLE; WXR OFF/failed or WX ON VD OFF -> NO WX DATA AVAILABLE; both -> the combined message.

A380X ECAM: WXR & TAWS and XPDR & TCAS lines sensed

ADR / IR / TCAS fault procedures tick from the real system selection

Branch	feature/a380x-checklists-surv
Aircraft	A380X
Commits	tip 14e842d8f (PR 6)
Base	feature/a380x-surv-panel
Status	In develop. PR draft 06.

Specification

- ADR 1+2 / 1+3 / 2+3 and IR 1+2 / 1+3 / 2+3 FAULT: the WXR & TAWS and XPDR & TCAS lines were hard-coded done; they tick from `L:A32NX_WXR_TAWS_SYS_SELECTED` / `L:A32NX TRANSPONDER_SYSTEM`.
- TCAS 1 / 2 FAULT: the XPDR & TCAS line is shown and ticks once the healthy system is selected.
- ECAM LINE-UP: CABIN CREW ... ADVISED (FCOM BEFORE TAKEOFF C/L). No surveillance lines in the normal checklists (FCOM PRO-NOR-C-L has none).

PR description (from the PR draft)

From `pr-drafts/06-a380x-checklists.md`, section "Summary of Changes" onwards.

Summary of Changes

ECAM LINE-UP checklist: gets the `CABIN CREW ... ADVISED` line that the EFB checklist already had (the FCOM's BEFORE TAKEOFF checklist has `CABIN CREW ... ADVISE`). The sensed-items arrays of the ECAM checklists are index-aligned with the items, so they are extended in step.

ECAM procedures: the ADR 1+2 / 1+3 / 2+3 FAULT and IR 1+2 / 1+3 / 2+3 FAULT procedures ask the crew to switch the WXR & TAWS and the XPDR & TCAS to SYS 1 or SYS 2, but both lines were hard-coded as always done. They now tick from the real selection (`L:A32NX_WXR_TAWS_SYS_SELECTED`, `L:A32NX TRANSPONDER_SYSTEM`), which the MFD STATUS & SWITCHING page and the pedestal buttons drive. The TCAS 1 / 2 FAULT procedures had their XPDR & TCAS line hidden behind a "replace with SURV SYS logic once implemented" TODO: it is now shown and ticks once the healthy system is selected.

Cockpit API Changes

No new LVars. Reads `L:A32NX_WXR_TAWS_SYS_SELECTED` and `L:A32NX TRANSPONDER_SYSTEM` (introduced by the SURV PR).

Screenshots (if necessary)

TO ADD: an ADR FAULT procedure with the XPDR & TCAS line before (always ticked) and after; the ECAM LINE-UP checklist with the CABIN CREW line.

References

- A380 FCOM PRO-NOR-C-L Normal checklists (BEFORE TAKEOFF: `CABIN CREW ... ADVISE`; no surveillance lines in any normal checklist).
- TO ADD:** the A380 FCOM abnormal procedures ADR 1+2 / 1+3 / 2+3 FAULT, IR ... FAULT and TCAS 1 / 2 FAULT (the WXR & TAWS ... SYS 1/2 and XPDR & TCAS ... SYS 1/2 lines), with page screenshots.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

- LINE-UP (ECAM): the CABIN CREW line exists and matches the EFB.
- Fail ADR 1+2 (or IR 1+2, and the other pairs): the WXR & TAWS and XPDR & TCAS lines tick only after selecting the system the procedure asks for.
- Fail TCAS 1 (and TCAS 2): the XPDR & TCAS line is shown and ticks once the healthy system is selected.

TCAS: RA-level intruders as TAs while RAs are inhibited

TA ONLY / low-altitude behaviour with the real hysteresis

Branch	fix/tcas-ta-only-intruder-level
Aircraft	A32NX and A380X
Commits	4 commits, tip 2518ca61e (PR 7)
Base	upstream master 2baa2b35e
Status	In develop. PR draft 07.

Specification

- With every RA inhibited (TA ONLY, or low altitude) an RA-level intruder is capped to TA: amber circle and TRAFFIC aural (FCOM: "All RAs are inhibited and converted into TAs"). Partial inhibits unchanged.
- Low-altitude inhibitions with hysteresis: RAs below 900 ft AGL descending / 1100 ft climbing, TA aural below 400 / 600 ft (A380 FCTM SI TCAS). Constants in `TcasConstants.ts`.
- A32NX ND: TA ONLY centred at the bottom (A320 FCOM DSC-34-SURV-60-20 figure).

PR description (from the PR draft)

From `pr-drafts/07-tcas-ta-only.md`, section "Summary of Changes" onwards.

Summary of Changes

With every RA inhibited (TA ONLY mode, or below 1000 ft radio altitude, the two cases that put up the "TA ONLY" message) the intruder classification still went to RA for an intruder that met the RA criteria. On the ND that intruder was drawn as a red RA square, and since it was not a TA either, it did not raise the TA state or the "traffic, traffic" aural. A real TCAS in TA ONLY shows it as an amber TA circle with the TA aural.

The desired intrusion level is now capped to TA while `Inhibit.ALL_RA` or `Inhibit.ALL_RA_AURAL_TA` is active. The partial inhibits (no climb / descend / increase descent RA) still issue RAs and are unchanged, and an RA that is already active keeps running until its own end conditions. Same change in the A32NX `TcasComputer` and the A380X `LegacyTcasComputer`, whose classification code is identical.

Found while comparing the ND traffic symbols with the reference symbology (hollow diamond other traffic, filled diamond proximate, amber circle TA, red square RA), which otherwise matches.

Cockpit API Changes

None.

Screenshots (if necessary)

TO ADD: the ND in TA ONLY with an intruder that meets the RA criteria, before (red square) and after (amber circle).

Update 2026-09-22 (uncommitted, second commit of this PR): low-altitude inhibition hysteresis

`updateInhibitions()` of both computers inhibited every RA below a flat 1 000 ft radio altitude and the TA aural below 500 ft. The manuals give a hysteresis: RAs are inhibited below 900 ft AGL in descent and 1 100 ft AGL in climb (A380 FCTM, Supplementary Information / TCAS; the FCOM's "below 1 000 ft +-100 ft"), the TA aural below 400 ft in descent and 600 ft in climb. Both are now state-based thresholds (`raInhibitedLow`, `taAuralInhibitedLow`), reset while the radio altitude is invalid. Suggested title: `fix(tcas): low-altitude RA and TA inhibitions with the real hysteresis`.

References

- IVAO training wiki, Traffic collision avoidance system (TCAS):
https://wiki.ivao.aero/en/home/training/documentation/Traffic_collision_avoidance_system-TCAS (traffic symbols and colours; used to check the ND symbology).
- A320 FCOM DSC-34-SURV-60-10-10, TA ONLY mode: "All RAs are inhibited and converted into TAs", "TA threshold is set to TAU 20 s", "'TA ONLY' is displayed on the NDs"; automatic TA ONLY below 1 000 ft AGL (and with windshear / stall / GPWS alerts).
- A380 FCOM DSC-34-20-40-10, TA ONLY mode: "All RAs are inhibited, and become TAs", automatic "below 1 000 ft +-100 ft".
- A380 FCTM SI TCAS (page 345 of the scanned FCTM): RAs inhibited below 900 ft AGL in descent / 1 100 ft AGL in climb, TAs below 400 / 600 ft AGL.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

- A32NX and A380X: TCAS in TA ONLY. Bring an intruder (AI traffic, slew or MSFS multiplayer) on a converging course that meets the RA criteria: it is an amber circle with the "TRAFFIC, TRAFFIC" aural, not a red square.
- Below 1000 ft radio altitude (the "TA ONLY" message shows): same.
- TCAS in TA/RA above 1000 ft: the same intruder still gives an RA (red square + RA aural). An RA in progress is not cut short.

4. The partial inhibits (climb inhibit at high weight/altitude, descend inhibit near the ground) still allow the other RA senses.

Independent CPT and F/O MCDUs

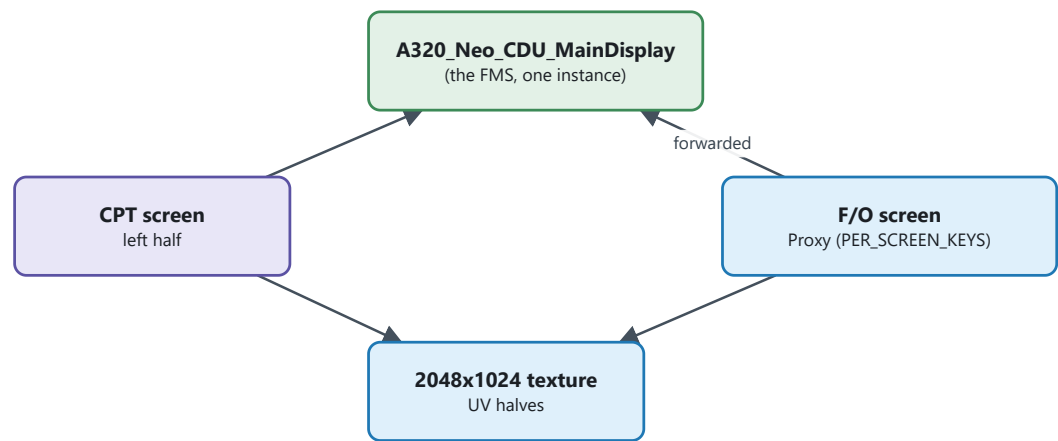
Two MCDU screens on one FMS

Branch	feature/a32nx-mcdu-independent
Aircraft	A32NX
Commits	5be670f87 , 62f3f2917 (PR 8)
Base	upstream master 2baa2b35e
Status	In develop. User-confirmed in-sim. PR draft 08; needs an aircraft-large-files PR.

Specification

- One MCDU gauge, 2048x1024 texture; each screen mesh samples one half (UV change in LOD00 / LOD01).
- The F/O screen is a Proxy over the single FMS: per-screen fields (page, LSK callbacks, scratchpad, messages, keys, DOM) are copied, everything else forwarded. The 88 legacy pages are unchanged.
- F/O keys to the F/O screen (AC 2), keyboard entry to the clicked screen, SimBridge remote MCDU shows both sides, ATSU status on both.

Component added or changed by the feature MSFS / simulator Aircraft system (systems host, FMS, FWS) Display / page
Data, LVar, storage Crew, flypad, external program



Architecture

The LOD00 model change must go to flybywiresim/aircraft-large-files first, then the submodule pointer is bumped in this PR.

PR description (from the PR draft)

From `pr-drafts/08-mcd�-independent.md`, section "Summary of Changes" onwards.

Summary of Changes

The two MCDU screens showed the same thing: there was one MCDU gauge on one texture and both screen meshes sampled all of it. This makes them independent without running the FMS twice. `A320_Neo_CDU_MainDisplay` is the aircraft's FMS (flight plan, guidance, EFIS symbols) as well as the MCDU display, and only one of it may exist. So both MCDUs are now two screens on that one FMS, the way the A380X does its two MFD screens (one gauge, one texture, both screens drawn into it):

- **Gauge and model:** the MCDU gauge is 2048x1024 and the two screen meshes each sample one half of it (the U of `SCREEN_MCDUL` runs 0..0.5, that of `SCREEN_MCDUR` 0.5..1). The CPT MCDU is drawn in the left half, the F/O MCDU in the right half. Only 4 UV values per LOD change (LOD01 in this repo, LOD00 in `aircraft-large-files`).
- **F/O screen:** a Proxy over the FMS instance. The fields that belong to one screen (page being shown, LSK callbacks, scratchpad, message queue, key handlers, DOM elements) are listed in `PER_SCREEN_KEYS` and the F/O screen has its own copy, everything else is forwarded to the FMS. Functions run with the proxy as `this`, so an entry made on the F/O MCDU puts its pages, scratchpad messages and dialogs on the F/O screen. The CPT screen stays the real instance with its code path unchanged. The 88 legacy pages are unchanged (they already take the MCDU through `LegacyFmsPageInterface`).
- **Keys and power:** the events of the F/O keys go to the F/O screen, checked against AC bus 2 (the CPT MCDU stays on AC ESS SHED).
- **Messages:** FMGC messages are queued on both screens and clearing one with CLR only clears it on that screen. What the FMS does by itself (request lights, PROG/PERF/IDENT page changes at a flight phase change, engine-out page request, INIT B to FUEL PRED) applies to every screen.
- **Keyboard entry, remote MCDU, ATSU:** the keyboard entry goes to the screen that was clicked and only one has it at a time (the pushed-key animation is on that MCDU only); the SimBridge remote MCDU shows the CPT screen on its left side and the F/O screen on its right side; ATSU system status that is not the answer to an entry is shown on the ATSU scratchpad of every MCDU.

Not changed (same as the CPT MCDU today): the MCDUs do not go back to the MENU page after a power loss.

Cockpit API Changes

None. The MCDU key events (`H:A320_Neo_CDU_{1,2}_BTN_*`) are unchanged, the F/O ones now reach the F/O screen. `panel.cfg: vCockpit08 (MCDU)` is 2048x1024.

Screenshots (if necessary)

TO ADD: before (both MCDUs mirrored) and after (F-PLN on one, PERF on the other), same camera angle; the two scratchpads with different text.

References

TO ADD - ONLY the Honeywell Pegasus Step 1A (Rev 0), 2009 manual (the template requires it for MCDU PRs): the pages showing that the two MCDUs have their own display, page and scratchpad while sharing the FMGC data. Ask on Discord for the manual if you do not have it. The A380X `panel.cfg` comment on its MFD texture is the precedent for the one-texture, two-screens technique. (For the issue text, not the PR: A320 FCOM DSC-22_10-10 "Two MCDUs are installed on the pedestal for flight crew loading and display of data".)

Additional context

A second gauge instance for the F/O MCDU (with template / `PART_ID` changes in the model behaviour XML) was tried and does not work in MSFS: the second gauge never reached its screen and two of the behaviour changes blanked both screens. The one-texture approach avoids the screen routing question and needs no second FMS, and because

both screens share one FMS nothing has to be synchronised between gauges. An earlier upstream attempt at the same idea, flybywiresim/a32nx#5383, was closed in 2022 (merge conflicts), its model change was never made; this PR makes it.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. CPT MCDU: MENU -> FMGC, INIT, F-PLN, PERF, PROG, DATA, typing and CLR, LSK entries, messages: exactly as before.
2. F/O MCDU shows the MCDU MENU and works by itself: put F-PLN on one MCDU and PERF on the other, they stay separate.
3. Scratchpads are independent (type on one, the other is unchanged).
4. FMS data is shared: an entry on one (flight plan, PERF values) shows on the other's page.
5. Messages: NOT ALLOWED / FORMAT ERROR appears only where you made the entry; FMGC queue messages (e.g. CHECK TAKE OFF DATA, ENTER DEST DATA) appear on both and CLR on one leaves the other.
6. BRT/DIM keys dim only their own screen. Remove AC 2: the F/O screen goes dark; the CPT stays on AC ESS SHED.
7. Keyboard entry (the MCDU keyboard input setting, `MCDU_KB_INPUT`): click the F/O screen, type in its scratchpad, then click the CPT screen: the keyboard moves and the key animations follow.
8. SimBridge remote MCDU: left and right show different content. ATSU: an ATIS request shows its status on the MCDU you used (and system status on both).

Radar, terrain and VD after the A380 / A320 manuals

VD cut along the flight plan, FCOM scale, terrain by true height, real-display look

Branch	feature/nd-manuals-pass
Aircraft	A32NX and A380X
Commits	15 commits, tip ddca9fe22 (PR 9)
Base	feature/nd-wxr-mode-label (chain top)
Status	In develop. Flown. PR draft 09.

Specification

Rule	Source	Implementation
VD cut along the active F-PLN in managed modes, along the track otherwise; width 2 x RNP	A380 FCOM DSC-31-20-40-10	L:A380X_VD_CUT_* LVars from EfiTawsBridge; gauge draws along it
Scale origin shows the mock-up and most of the trajectory	FCOM DSC-31-20-40-10	aircraft / path at 3/4 of the window
Terrain height under the mock-up geometrically correct, not baro dependent	A380 FCTM SI-70	range from the ADR baro altitude
No magenta on the VD; no VD weather without TERR	FCOM DSC-31-20-40-10	VD weather layer
Radar scans on the ground	A320 FCOM DSC-34-SURV-30-30	ground inhibit removed
MAP mode: black water, green ground, amber mountains	A320 FCOM WXR panel	terrain-view ground map
30 s buffer fill	A380 FCOM WXR operational recommendations	one slow sweep
Real antenna sweep (A320 RDR-4B 180 deg, A380 RDR-4000 160 deg)	vendor documents	cone
No weather above 320 NM	A380 FCOM DSC-31-20-50	range limit

The `ndwxr` module was split into nine files by concern (quality pass), zero warnings for both aircraft.

PR description (from the PR draft)

From `pr-drafts/09-a380x-vd-cut.md`, section "Summary of Changes" onwards.

Summary of Changes

- **Vertical cut (FCOM DSC-31-20-40-10):** `EfisTawsBridge` publishes the cut as `L:A380X_VD_CUT_*` LVars: along the active flight plan in the managed lateral modes (a vertex per path vector, the midpoint of each turn, a 160 NM straight extension, a 200 NM window from just behind the aircraft, twice a second), along the track (IR true track) in the selected modes, as the FCOM's "VIEW ALONG ACFT TRK". The native gauge draws the terrain profile piece by piece along it (falls back to the track when the aircraft is more than 5 NM off the plan), with a width of $2 \times \text{RNP}$ (1 NM terminal, 2 NM en route above 18 000 ft) as the highest terrain across five columns, and the grey area from the FMS's next track change > 3 deg. The weather on the VD follows the same cut with a zero width (beam-width sampling).
- **Scale (FCOM "the FMS defines the origin of the vertical scale in order to display the aircraft mock-up and most of the aircraft trajectory"):** with a trajectory the highest of aircraft / path sits at three quarters of the 4 deg window, the top is capped at 70 000 ft and the bottom may go below sea level, so the ground stays on the plot in cruise as on the real aircraft (photo at FL400). Without a trajectory the aircraft stays in the middle.
- **Terrain placement (FCOM / FCTM SI-70 "the height of the aircraft above the terrain displayed on the VD is geometrically correct and does not vary with the altimeter setting"):** the terrain view's range is set from the ADR baro altitude, so the gap under the mock-up is the true height whatever the baro setting. Water is drawn from a water-mask view in the same teal as the ND's terrain display (the FCOM says blue; the ND's colour is what the crew sees next to it). Magenta is not drawn on the VD (FCOM), and the VD weather is removed while TERR is unavailable (FCOM: "the VD does not display the weather when the TERR function is not available").
- **Weather picture (both aircraft, from the manuals pass):** the radar scans on the ground too (A320 FCOM DSC-34-SURV-30-30), MAP mode is a ground map from the terrain view, the picture fills over 30 s after the radar starts transmitting (A380 FCOM operational recommendations), the ARC picture keeps the ND's bottom corners free as on the real display, water on the terrain display is the real display's teal, and the TERR peaks box figures come from SimBridge over SimConnect (PR 4 draft). The A380's AUTO mode off-path display is NOT modelled: it was built from a third MapView in the engine's top-view radar mode and removed again, because the engine runs one radar per aircraft and that view changed what the beam views delivered (rain at the aircraft's level over the whole range instead of the cells; the A32NX next to it was clean). Both aircraft show the same picture.

Cockpit API Changes

New LVars, documented in `fbw-a380x/docs/a380-simvars.md`: `L:A380X_VD_CUT_MODE` (0 track / 1 plan), `L:A380X_VD_CUT_COUNT`, `L:A380X_VD_CUT_{i}_LAT` / `_LON` ($i < 32$, degrees), `L:A380X_VD_CUT_TRACK_CHANGE_NM`. Reads the IR true track words.

Screenshots (if necessary)

TO ADD: the VD through a turn in NAV (profile follows the route, grey area at the turn), the same in HDG (along the track), cruise at 160 NM (ground visible, aircraft in the upper quarter), an approach with terrain (gap under the mock-up with QNH vs STD).

References

- A380 FCOM DSC-31-20-40-10 (VD): vertical cut along the active flight plan / current track, cut width $2 \times \text{RNP}$, vertical scale origin, terrain by true height, water blue, magenta not displayed, weather needs TERR (PDF pages 2365-2394 of the KAL FCOM).
- A380 FCTM SI-70 Safety altitude and terrain / Vertical trajectory / Terrain (VD height geometrically correct; MORA/MSA along the F-PLN in NAV, along the track in HDG/TRACK; grey area = next turning point).
- Real A380 photo at FL400 (ground on the VD, aircraft at $\sim 75\%$ of the height).

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. NAV mode with a turn ahead, VD on: the terrain profile follows the route through the turn (compare with the ND terrain along the green line); a grey area starts at the turn. Switch to HDG: the profile is along the track; back to NAV: along the plan.
2. Cruise at 160 NM: the ground and the terrain profile are visible, the aircraft symbol in the upper quarter of the VD.
3. Approach with terrain nearby, QNH then STD: the gap under the aircraft symbol does not change.
4. Water on the cut: blue from sea level down. TERR SYS OFF: the VD weather goes with the terrain.

A380X MFD FMS and SURV pages to the FCOM

Every MFD FMS page laid out on its FCOM figure, the missing pages and functions

Branch	feature/a380/mcdu-improvements
Aircraft	A380X
Commits	8 own commits (on PR 6): 3800ff39d, e9d159981, ed33c1686, 752007449, fbef33990, edb07a695, 197a5f6aa, a09803fa8
Base	feature/a380x-checklists-surv
Status	In develop. Deployed and used by the user. PR draft 10.

Specification

- All 27 existing FMS pages audited against the A380 FCOM (DSC-22-FMS-20-30, PDF p.1009-1409): layouts rebuilt on the FCOM figures (absolute FCOM positions, border-box sizes as in Coherent GT), FCOM data entry formats (DSC-22-FMS-20-100).
- New pages: ALTERNATE, CLOSEST AIRPORTS, OFFSET, EQUI-TIME POINT, LL XING / TIME MKR, DATA / PRINTER, COMPANY WIND DATA REQUEST, COMPANY T.O DATA REQUEST, RECEIVED COMPANY T.O DATA, COMPANY F-PLN REPORT / REQUEST, REJECTED ATC INFO, TRANSFER TO MAILBOX.
- Functions: FMS print functions with printouts in the flypad; FUEL PLANNING computes and confirms the minimum BLOCK (FCOM P 179); PERF CRZ LRC; VERT REV RTA; FMS P/N panel; pilot-stored runways.
- SURV pages (CONTROLS, STATUS & SWITCHING) on DSC-34-20-60-50; SURV memos of the WXR and TAWS, TCAS STBY with ALT RPTG OFF.

Kept where FBW had a newer standard than the 2011 FCOM only when the user asked; the FCOM PDF is the reference for layout.

PR description (from the PR draft)

From `pr-drafts/10-a380x-mfd-fcom-pages.md`, section "Summary of Changes" onwards.

Summary of Changes

Every A380X MFD FMS page was compared with its figure in the A380 FCOM (DSC-22-FMS-20-30, PDF pages 1009-1409) and re-laid out on it, and the FCOM pages and functions FBW did not have were added. Each layout value and rule names its FCOM page in a code comment.

- **Layout:** the pages use the FCOM figure coordinates (`common/FcomLayout.tsx: fcomAt, fcomRight, fcomCentre, fcomLine`), the global FMS frame of the figures (title bar, 40 px entry fields, 22.5 px labels, 38 px tab bar), and box sizes counted border-box as Coherent GT renders them. Field contents are centred, dropdown arrows sit inside the frame, open menus are raised above the next items.
- **Entry formats:** the FCOM data entry formats (DSC-22-FMS-20-100) for every field, with the FCOM error messages.
- **New pages:** ALTERNATE, CLOSEST AIRPORTS, OFFSET, EQUI-TIME POINT, LL XING / TIME MKR, DATA / PRINTER, COMPANY WIND DATA REQUEST, COMPANY T.O DATA REQUEST, RECEIVED COMPANY T.O DATA (laid out; wired by PR 11), COMPANY F-PLN REQUEST / REPORT, REJECTED ATC INFO, TRANSFER TO MAILBOX.
- **Functions:** FMS print functions with the printouts shown in the flypad; FUEL PLANNING computes the minimum BLOCK from the predictions and CONFIRM BLOCK inserts it (FCOM P 179); PERF CRZ LRC (99 % of the maximum specific range, FCOM PER-IFT) and MAX TURB speed; VERT REV RTA with the time error thresholds; FMS P/N panel; pilot-stored runways; AIRWAYS with 31 segments and automatic airway connection; DATA / STATUS and AIRPORT behaviour per the FCOM.
- **SURV pages:** CONTROLS and STATUS & SWITCHING on DSC-34-20-60-50 P 4 / P 6 (XPDR AUTO / ON / STBY, TCAS and WXR option rules, ELEVN / TILT and GAIN fields); SURV memos of the WXR and TAWS; TCAS STBY with ALT RPTG OFF.

Where FBW had a newer FMS standard than this 2011 FCOM and the FCOM figure shows something else, the FCOM figure was followed.

Cockpit API Changes

New LVars documented in `fbw-a380x/docs/a380-simvars.md` for the SURV pages (WXR ELEVN / TILT / GAIN settings). No LVar removed.

Screenshots (if necessary)

TO ADD: each FMS page next to its FCOM figure (before / after).

References

- A380 FCOM DSC-22-FMS-20-30 (MFD FMS pages, PDF p.1009-1409), DSC-22-FMS-20-100 (data entry formats), DSC-34-20-60-50 (SURV pages).

Additional context

The pages were compared with the FCOM figures in a local browser harness that renders the MFD bundle with the sim's box model.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. Open every FMS page (ACTIVE, POSITION, SEC INDEX, DATA menus) and compare it with its FCOM figure.
2. Enter values in each field, including out-of-range and badly formatted entries: FCOM format and error messages.
3. PRINT functions: the printout appears in the flypad.

4. FUEL PLANNING with a flight plan, cruise FL, ZFW and ZFWCG and no BLOCK: the computed BLOCK appears, CONFIRM BLOCK inserts it.
5. SURV CONTROLS / STATUS & SWITCHING: every button and field.

A380 takeoff calculator and company T.O data

Flypad takeoff performance for the A380X, with the FMS company takeoff data exchange of the A380 FCOM

Branch	feature/a380/efb/take-off-calculator
Aircraft	A380X (flypad shared with the A32NX, see document 12)
Commits	c16e6e877 feat(a380x/efb): A380 takeoff calculator with company T.O data to the FMS
Base	feature/a380/mcdu-improvements (document 10)
Status	Merged in develop. Tested in the sim by the user.

Purpose

The A380X flypad had no takeoff calculator. The A32NX one uses FBW's A320 performance model, which does not exist for the A380. This feature adds an A380 calculator built from public Airbus data and the FCOM rules, a runway drawing that shows the distance needed on the runway, and the FCOM company takeoff data function that sends the results to the MFD, where the crew inserts them on PERF T.O.

Specification

Data sources

- **Airbus A380 Aircraft Characteristics** (AC, 1 Dec 2025), take-off weight limitation charts, PDF p.135-139 (ISA and ISA+15, Trent 900 and GP7200). The curves are vector paths: they were extracted exactly (not digitised from pictures): runway 1500-5500 m, TOW 300-650 t, airport altitudes 0/2000/4000/6000/8000 ft. FBW's A380X has Trent 972 engines, so the Trent 900 charts are used. The charts are "for information only": dry runway, no wind, no slope.
- Tables in `a380x_takeoff_data.ts`: 1700-5100 m x 0-8000 ft, ISA and ISA+15. Below 1700 m the chart curves start distorted: the table does not go lower, shorter runways are extrapolated (W proportional to L).
- A380 FCOM for the rules (table below). The FBW VMCA / VMCG tables are A320 placeholders: the FCOM gives no values.

Two data modes

- **Airbus data + estimates** (default): chart weights, plus FBW-model estimates for what the charts do not cover (wind, slope, temperatures between the ISA and ISA+15 curves, FLEX). Estimated values are amber with an asterisk, as on the OIS.
- **Airbus data only**: only what the charts give; anything else is an error message instead of an estimate.

Model

- MTOW(perf) from the chart for the runway length and airport altitude, corrected for temperature: $W \sim \sigma^k \times \text{thrust}^j$, with k fitted on ISA vs ISA+15 (0.45 short runway to 0.2 long) and j on the altitude curves (0.3-0.45).
- FLEX: $W(T) = W(\text{TREF}) \times (\text{density ratio})^k \times \text{thrust ratio}$, thrust ratio linear from 1 at TREF to 0.6 at TMAXFLEX (FCOM: 40 % maximum reduction). Checked on the FCOM example (560 t on a 614 t MTOW -> FLEX 38 degC; model 37).
- Speeds: $V_2 \geq 1.13 \text{ VS1G}$ and $\geq 1.10 \text{ VMCA}$, $V_R \geq 1.05 \text{ VMCA}$, $V_1 \geq \text{VMCG}$; V_R/V_2 and V_1/V_R ratios from the FCOM RESULTS example (V_1 153 / V_R 168 / V_2 174 at 560 t).
- Wind: regulatory 50 % of the headwind, 150 % of the tailwind. Line-up allowances 172.5 / 219.8 ft (90 / 180 deg entry).
- **Takeoff run on the runway**: the required length is the runway length at which the chart weight limit equals the TOW (bisection). V_1 , V_R and 35 ft points: $\text{TOD0} = \text{required} / 1.15$ (CS 25.113), constant acceleration to $V_2 + 10$.

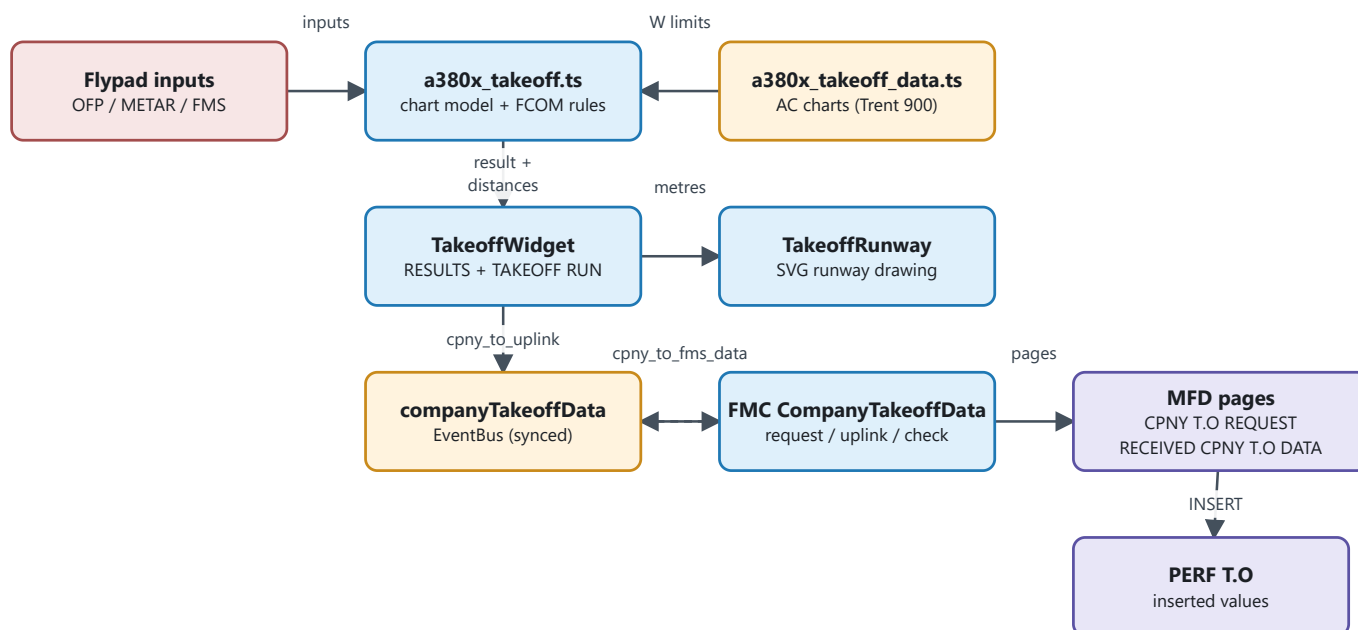
Screen

- Four input sections: RUNWAY (ident or manual TORA, heading, elevation, slope, entry angle), CONDITIONS (runway condition, wind, OAT, QNH, anti-ice, packs), AIRCRAFT (TOW, T.O CG, flaps, thrust), DEPARTURE (THR RED / ACCEL /

EO ACCEL, defaulted to the elevation + the flypad CONFIG altitudes; noise abatement).

- Fill from: SimBrief OFP, METAR, or **FMS** (the MFD answers with its takeoff data, including the runway conditions of the last SEND T.O REQUEST).
- RESULTS panel in the layout of the OIS takeoff application (FCOM PER-TOF-TOR-SRS P 1 figure): TOW, MTOW(perf), V1/VR/V2, margin, FLEX or TOGA, flaps, T.O CG, T.O SHIFT, THR RED / ACCEL / EO ACCEL.
- TAKEOFF RUN panel: runway drawing (length, shift, line-up, V1 / VR / 35 ft marks, required distance bar, margin), TOGA or any FLEX from the lowest allowed to the maximum with +/- buttons and a slider, whatever thrust was chosen first (FCOM: a lower FLEX may be used to increase the margin).

■ Component added or changed by the feature ■ MSFS / simulator ■ Aircraft system (systems host, FMS, FWS) ■ Display / page
■ Data, LVar, storage ■ Crew, flypad, external program



Architecture: calculator, flypad screen and the company T.O data path to the MFD

FCOM rules implemented

Rule	Source	Implementation
TREF = ISA+15, TMAXFLEX = ISA+60, max 40 % thrust reduction, no FLEX on a contaminated runway	FCOM PER-TOF-THR-FLX	FLEX range and thrust ratio; TOGA forced on contaminated runways
Range of TFLEX: any FLEX from max(TREF, OAT) to the maximum may be used	FCOM PER-TOF-THR-FLX	FLEX slider in the TAKEOFF RUN panel
V2 >= 1.10 VMCA, >= 1.13 VS1G; VR >= V1, >= 1.05 VMCA; V1 >= VMCG	FCOM PER-TOF-TOR-SRS P 1-2	speed limits
RESULTS panel layout	FCOM PER-TOF-TOR-SRS P 1 (PDF p.6609)	ResultsPanel
Uplink with or without request; message COMPANY T.O DATA RECEIVED WAITING FOR INSERTION	FCOM DSC-22-FMS-20-30	FMC CompanyTakeoffData
Up to 4 runways x 2 thrust ratings received	FCOM DSC-22-FMS-20-30	uplink list
Runway amber when not the active one; INSERT only with the active runway	FCOM DSC-22-FMS-20-30 P 312-320	RECEIVED page + check

TOW margin -2 t / +7 t of FUEL&LOAD, else amber + UPLINK/ACTIVE TOW DISAGREE	FCOM DSC-22-FMS-20-30	check before INSERT
INSERT needs speeds, TOW, T.O CG, wind, runway condition, baro, temperature	FCOM DSC-22-FMS-20-30	INSERT enabling
INSERT puts speeds, flaps, thrust, shift, THR RED / ACCEL / EO ACCEL, noise (not the CG: the crew enters it in THS FOR)	FCOM + SOP	insertCompanyTakeoffData
T.O LIMIT = remaining runway length (FT), T.O SHIFT in M; TEMP dashes for FLEX/DERATED	FCOM DSC-22-FMS-20-30	RECEIVED page fields
SEND T.O REQUEST -> REQUEST PENDING...; no answer in 4 min -> NO COMPANY REPLY	FCOM DSC-22-FMS-20-30	request timer
Required takeoff distance = 1.15 x TOD0 (all engines)	CS 25.113	V1 / VR / 35 ft marks

Interfaces

Item	Type	Meaning
cpny_to_fms_data	EventBus (synced)	FMS takeoff data / request, answers a flypad import
cpny_to_fms_data_request	EventBus	flypad asks the FMS for its data
cpny_to_uplink	EventBus	takeoff data sent by the flypad (one per runway and thrust)

Verification

- MFD and flypad checked in a local browser harness (received page fields, INSERT populating PERF T.O).
- Model checked against the FCOM example; unit-level checks in a Node harness.
- User in-sim test: accepted.

Limitations

- The charts are dry / no wind / no slope: every correction is an FBW estimate, flagged amber.
- VMCA / VMCG are the FBW A320 placeholders (no A380 values in the FCOM).
- Below 1700 m the required length is extrapolated (real-data mode: "less than 1700 m").

PR description (from the PR draft)

From `pr-drafts/11-a380-takeoff-calculator.md`, section "Summary of Changes" onwards.

Summary of Changes

The A380X flypad gets a takeoff calculator, and the MFD the FCOM company takeoff data function that receives its results.

Calculator (`a380x_takeoff.ts`, `a380x_takeoff_data.ts`)

- Weight limits from the Airbus A380 Aircraft Characteristics (1 Dec 2025), take-off weight limitation charts, Trent 900, ISA and ISA+15 (PDF p.135-139). The curves are vector paths and were extracted exactly: runway 1700-5100 m, airport altitude 0-8000 ft.
- Corrections the charts do not give are FBW-model estimates, amber with an asterisk: temperature ($W \sim \sigma^k \times \text{thrust}^j$, fitted on the charts), FLEX (thrust ratio 1 at TREF to 0.6 at TMAXFLEX, FCOM 40 % maximum; checked on the FCOM example 560 t / 614 t -> FLEX 38), wind (50 % headwind / 150 % tailwind), slope, line-up (172.5 / 219.8 ft). A "Data" select switches to Airbus data only.
- Speeds: $V2 \geq 1.13 VS1G$ and $\geq 1.10 VMCA$, $VR \geq 1.05 VMCA$, $V1 \geq VMCG$, ratios from the FCOM RESULTS example.
- Takeoff run: the required length is where the chart limit equals the TOW (bisection); $V1 / VR / 35$ ft points from $TOD0 = \text{required} / 1.15$ (CS 25.113) with a constant acceleration to $V2 + 10$.

Flypad screen: RUNWAY / CONDITIONS / AIRCRAFT / DEPARTURE inputs (fill from the OFP, the METAR or the FMS), the RESULTS panel of the OIS takeoff application (FCOM PER-TOF-TOR-SRS P 1), and a TAKEOFF RUN panel with a runway drawing (shift, line-up, $V1 / VR / 35$ ft marks, required distance, margin) for TOGA or any FLEX of the allowed range (FCOM PER-TOF-THR-FLX, range of TFLEX).

Company T.O data (FCOM DSC-22-FMS-20-30): SEND T.O REQUEST (REQUEST PENDING..., NO COMPANY REPLY after 4 min), uplinks for up to 4 runways x 2 thrust ratings, message COMPANY T.O DATA RECEIVED WAITING FOR INSERTION, INIT / PERF button RECEIVED CPNY T.O, the RECEIVED COMPANY T.O DATA page (runway amber when not the active one, TOW amber outside -2 t / +7 t with UPLINK/ACTIVE TOW DISAGREE, TEMP dashes for FLEX, T.O LIMIT and T.O SHIFT) and INSERT, which fills PERF T.O (speeds, flaps, thrust, shift, THR RED / ACCEL / EO ACCEL, noise; not the CG, which the crew enters in THS FOR).

Cockpit API Changes

None (EventBus topics `cpny_to_fms_data`, `cpny_to_fms_data_request`, `cpny_to_uplink`, synced between the instruments).

Screenshots (if necessary)

TO ADD: the flypad screen (TOGA and FLEX runs), the RECEIVED COMPANY T.O DATA page, PERF T.O after INSERT.

References

- Airbus A380 Aircraft Characteristics, Airport and Maintenance Planning (1 Dec 2025), take-off weight limitation charts.
- A380 FCOM PER-TOF-THR-FLX, PER-TOF-TOR-SRS P 1-2, DSC-22-FMS-20-30 P 312-320 (RECEIVED COMPANY T.O DATA). CS 25.113.

Additional context

The FBW VMCA / VMCG tables are A320 placeholders (the FCOM gives no values). The charts are dry, no wind, no slope: every other case is an estimate and flagged as one.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. Fill from the OFP, calculate: results and takeoff run; switch TOGA / FLEX and move the FLEX slider: the distances change.
2. Data select "Airbus data only": corrections not covered by the charts give an error instead of an estimate.
3. SEND TO FMS, then MFD RECEIVED CPNY T.O: the data per runway / thrust; INSERT fills PERF T.O.
4. Change the MFD runway or the ZFW: the runway / TOW turns amber and INSERT is not available.
5. MFD SEND T.O REQUEST, then Fill from FMS in the flypad: the request conditions come in.

A320 takeoff calculator and MCDU uplink takeoff data

The A380 flypad screen for the A32NX, with the A320 FCOM UPLINK TO DATA pages on the MCDU

Branch	feature/a32nx/efb/take-off-calculator
Aircraft	A32NX (and the shared flypad screen of both aircraft)
Commits	b25f790a2 feat(a32nx/efb): A320 takeoff calculator with the MCDU uplink takeoff data
Base	feature/a380/efb/take-off-calculator (document 11)
Status	Merged in develop. Deployed; MCDU pages not yet flown.

Purpose

The same calculator screen for the A320, fed by FBW's A320 performance model, and the takeoff data link of the A320 FCOM between the flypad (the "ground station") and the MCDU.

Specification

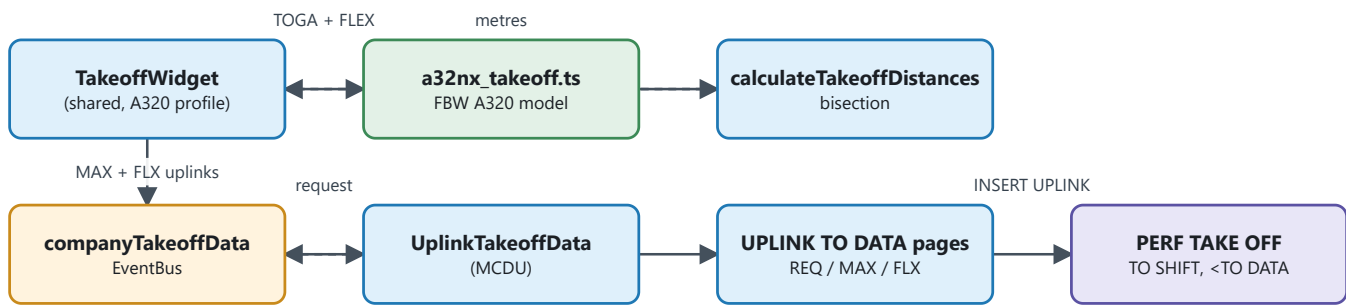
Flypad (shared screen)

- One screen for both aircraft (`TakeoffWidget`, `TakeoffRunway`, `TakeoffFmsLink`); the differences are one profile per aircraft (FMS TOW margins, noise fields and T.O CG only on the A380, text keys).
- Distances from the FBW A320 model: the required length by bisection on the runway length (1000-5000 m), re-running the model; for a FLEX, the length where the model still allows that FLEX.
- The FBW A320 model gives different speeds for TOGA: a separate TOGA calculation feeds the TOGA run and the MAX uplink.
- Performance at forward CG below 27 % MAC (A320 FCOM PER-LOD: forward-CG performance for a CG below 27 %).
- THS shown in the results; V1 / VR / 35 ft estimates share one helper with the A380 (standard atmosphere TAS).
- SEND TO FMS sends two uplinks: MAX (TOGA) and FLEX.

MCDU (A320 FCOM DSC-22_20-50-10-28 P 89-99, DSC-22_45)

- PERF TAKE OFF: `<TO DATA` (6L) opens UPLINK TO DATA REQ, only in the PREFLIGHT and DONE phases on the active plan; the TO SHIFT field works (1 m to the runway length, CLR).
- UPLINK TO DATA REQ (two runways): conditions of the request, sends it; the flypad answers.
- UPLINK MAX TO DATA / UPLINK FLX TO DATA (up to four runways, CONTAM scroll): the received data per runway.
- INSERT UPLINK: only with the runway of the flight plan, and a TOW not more than 1 t below or 3 t above the FMS TOW; inserts V1/VR/V2, FLAPS/THS, FLEX (cleared for MAX), THR RED/ACC, ENG OUT ACC and the shift.
- Messages: TAKEOFF DATA UPLINK, INVALID TAKEOFF UPLINK, REQUEST IS PENDING, NO ANSWER TO REQUEST after 4 min.

 Component added or changed by the feature  MSFS / simulator  Aircraft system (systems host, FMS, FWS)  Display / page
 Data, LVar, storage  Crew, flypad, external program



Architecture

FCOM rules implemented

Rule	Source	Implementation
UPLINK TO DATA REQ for two runways, data received for four	A320 FCOM DSC-22_20-50-10-28 P 89-92	pages + UplinkTakeoffData
Not insertable if the uplink TOW is 3 t greater or 1 t lower than the FMS TOW; runway must match	P 93-96	INSERT UPLINK check
TO SHIFT positive, not greater than the runway length	P 99 [2R]	PERF TAKE OFF field
UPLINK TO DATA prompt in PREFLIGHT and DONE	P 99 [6L]	
TAKEOFF DATA UPLINK / INVALID TAKEOFF UPLINK messages	DSC-22_45	NXSystemMessages
Forward-CG performance below 27 % MAC	A320 FCOM PER-LOD	calculator input

Verification

- tsc, ESLint, Prettier clean; A32NX MCDU and EFB, A380X EFB and MFD build.
- Flypad checked in a headless harness: 70 t on 2500 m gives TOGA 1728 m, FLEX 59 2444 m; labels do not overlap.
- Deployed; the MCDU UPLINK pages have not been seen on screen yet.

PR description (from the PR draft)

From `pr-drafts/12-a320-takeoff-calculator.md`, section "Summary of Changes" onwards.

Summary of Changes

The flypad takeoff screen of PR 11 is shared by both aircraft (`TakeoffWidget`, `TakeoffRunway`, `TakeoffFmsLink`); what differs per aircraft is one profile (FMS TOW margins, noise fields and T.O CG only on the A380, text keys). This PR also completes the takeoff screen and the A380 calculator of PR 11 (see "Both aircraft" and "A380 calculator" below).

A32NX calculator

- Takeoff run from the FBW A320 performance model: the required length by bisection on the runway length (1000-5000 m); for a FLEX, the length where the model still allows that temperature.
- The FBW A320 model gives other speeds for TOGA: a separate TOGA calculation feeds the TOGA run and the MAX uplink.
- Performance at forward CG below 27 % MAC (A320 FCOM PER-LOD). THS in the results.
- SEND TO FMS sends the takeoff data of the thrust selected for the takeoff run only (MAX, or FLEX at its temperature, with that thrust's own speeds).

MCDU (A320 FCOM DSC-22_20-50-10-28 P 89-99, DSC-22_45)

- PERF TAKE OFF: `<TO DATA` (6L) in PREFLIGHT and DONE; the TO SHIFT field (1 m to the runway length, CLR).
- UPLINK TO DATA REQ (two runways), UPLINK MAX TO DATA / UPLINK FLX TO DATA (four runways, CONTAM scroll).
- INSERT UPLINK: runway of the flight plan only, TOW not more than 1 t below / 3 t above the FMS TOW; inserts V1 / VR / V2, FLAPS / THS, FLEX (cleared for MAX), THR RED / ACC, ENG OUT ACC, shift.
- Messages TAKEOFF DATA UPLINK, INVALID TAKEOFF UPLINK, REQUEST IS PENDING, NO ANSWER TO REQUEST (4 min).
- No INSERT UPLINK prompt for an incomplete uplink, e.g. without V1 or VR (A320 FCOM DSC-22_20-70 P 6).

Both aircraft (takeoff screen of PR 11)

- One company takeoff data link for the A380 and A320 FMS (`CompanyTakeoffDataLink`, fbw-common): the request and its 4 min timeout, the data of up to four runways, the common checks and the answer to the flypad import; each aircraft adds its own checks.
- The Data selector (Airbus data only) is removed: with the charts alone the A380 calculator gives no FLEX, V1 or VR and refuses any wind, slope or temperature above TREF. The estimates are always used and stay flagged (amber, asterisk).
- The labels of the missing inputs (heading, TORA, elevation, slope, wind, OAT, QNH, TOW) are amber, so the pilot sees why CALCULATE is not available.
- OFP import: reads the current SimBrief OFP when the flypad has none, handles an OFP without METAR and a runway that is not found, and reports every failure.

A380 calculator (PR 11)

- A TOW of exactly 510 t (the structural MTOW) no longer gives CG OUT OF LIMITS: the envelope limits are inclusive, as on the A320.
- 42 unit tests against the Airbus A380 AC 3-3 chart points and the A380 FCOM (PER-TOF-TOC corrections, PER-TOF-TOR-SRS minimum speeds, PER-TOF-THR-FLX FLEX range, LIM-12 limits).
- The A380X EFB is rebuilt when the calculator changes (EFB `extraDeps`).

Cockpit API Changes

None.

Screenshots (if necessary)

TO ADD: the flypad screen on the A32NX (TOGA and FLEX), the three MCDU pages, PERF TAKE OFF after INSERT UPLINK.

References

- A320 FCOM DSC-22_20-50-10-28 P 89-99 (UPLINK TO DATA pages, PERF TAKE OFF), DSC-22_20-70 P 6 (INSERT UPLINK not displayed for an incomplete uplink), DSC-22_45 (takeoff data function), PER-LOD.
- A380 FCOM PER-TOF-TOC, PER-TOF-TOR-SRS, PER-TOF-THR-FLX, LIM-12; Airbus A380 Aircraft Characteristics (Dec 01/25) 3-3 (tests).

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. Flypad on the A32NX: calculate, TOGA and FLEX runs; 70 t on 2500 m gives about 1730 m at TOGA.
2. SEND TO FMS; MCDU PERF TAKE OFF `<TO DATA`, then the MAX and FLX pages: INSERT UPLINK fills PERF TAKE OFF.
3. Another runway in the flight plan, or a ZFW 4 t higher: INSERT UPLINK refused.
4. TO SHIFT entry and CLR.
5. SEND TO FMS with TOGA selected, then with a FLEX: only the selected thrust reaches the MCDU (MAX or FLX page).
6. Clear an input (e.g. OAT): its label turns amber and CALCULATE is not available.
7. A380X: TOW 510.0 t with a CG inside the envelope calculates (was CG OUT OF LIMITS); no Data selector on the screen.
8. `npx vitest run fbw-a380x/src/systems/shared/src/performance/a380x_takeoff.spec.ts` (42 tests).

Pilot stored elements kept across sessions

Optional flypad setting that keeps the MFD pilot-stored waypoints, nav aids, routes and runways

Branch	feature/a380/mfd/pilot-stored-elements-persistence
Aircraft	A380X (setting shared)
Commits	7636abc1c feat(a380x/mfd): keep the pilot stored elements across sim sessions (flypad setting)
Base	pr/10-a380x-mfd-fcom-pages (first built on the old fork master 280800cf7)
Status	Merged in develop. Harness-tested over two sessions.

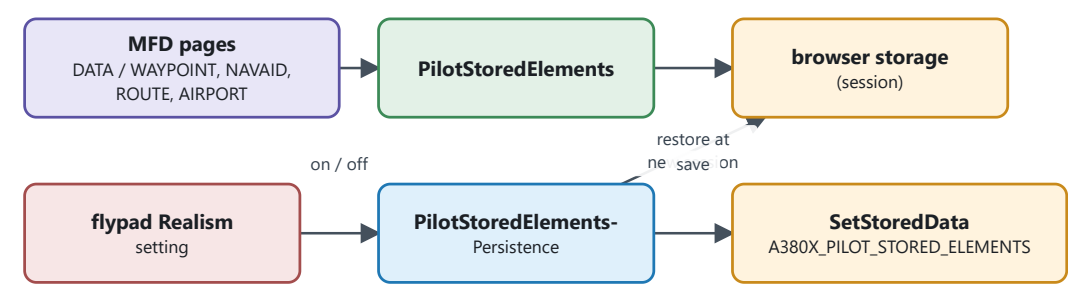
Purpose

Stored waypoints, nav aids, routes and runways disappeared when the sim closed. The FCOM says "All the pilot-stored elements are deleted when all FMCs are shut down" (DSC-22-FMS), so that stays the default; the user chose an optional flypad setting to keep them.

Specification

- Realism page setting **Keep pilot stored elements** (off by default = FCOM behaviour, marked unrealistic).
- While on: the browser-storage keys of the stored elements (`A32NX.StoredWaypoints` , `A380X.PilotStored{Nav aids, Routes, Runways}`) are saved to the persistent storage (`SetStoredData A380X_PILOT_STORED_ELEMENTS`).
- At a new session (sentinel key `A380X.PilotStoredElementsSession`) they are restored before the FMS loads them.

Component added or changed by the feature MSFS / simulator Aircraft system (systems host, FMS, FWS) Display / page Data, LVar, storage Crew, flypad, external program



Architecture

Rules

Rule	Source	Implementation
Pilot-stored elements deleted when all FMCs are shut down	A380 FCOM DSC-22-FMS	default behaviour kept (setting off)

PR description (from the PR draft)

From `pr-drafts/13-pilot-stored-elements-persistence.md`, section "Summary of Changes" onwards.

Summary of Changes

The A380 FCOM says "All the pilot-stored elements are deleted when all FMCs are shut down" (DSC-22-FMS); that stays the default. A new flypad realism setting, **Keep pilot stored elements** (off by default, marked unrealistic), saves the stored waypoints, nav aids, routes and runways in the persistent storage (`SetStoredData A380X_PILOT_STORED_ELEMENTS`) and restores them at the next session, before the FMS loads them (a session sentinel tells a new session from a reload).

Also in this PR:

- The OIT flyPad (OITLegacy) gets the new realism option too, with the A380X EFB value; without it the `FBW_TYPECHECK` build fails.
- 17 unit tests (`PilotStoredElements.spec.ts`): the FCOM database limits (20 NAVAIDs and 10 runways, the first created one deleted when full; 5 routes, then PILOT RTEs LIST FULL), replacement instead of duplicates, city pair routes, deletions, the NAVAID class fields, unreadable storage, and the setting across sessions (kept or deleted, written only on a change, not restored twice in a session, the kept copy dropped when the setting is turned off).
- Test setup fix (`fbw-common/src/jest/setupJestMock.ts`, test only): Node 22+ defines its own `localStorage` global, undefined without `--localstorage-file`, and vitest keeps it instead of the jsdom one. The setup now uses the jsdom storage, or an in-memory one, when `localStorage` is undefined. The new tests need it, and it also makes the existing `WaypointEntryUtils.spec.ts` and `SimBriefUplinkAdapter.spec.ts` load again on Node 22+ (they failed to load in the dev container, Node 26).

Cockpit API Changes

None. New stored data key `A380X_PILOT_STORED_ELEMENTS`, new setting `KEEP_PILOT_STORED_ELEMENTS`.

Screenshots (if necessary)

TO ADD: the Realism setting; DATA / WAYPOINT after a sim restart with the setting on.

References

- A380 FCOM DSC-22-FMS (pilot-stored elements deleted at FMC shutdown).

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. Setting off: store a waypoint, restart the sim: it is gone (FCOM).
2. Setting on: store waypoints, nav aids, a route and a runway, restart the sim: they are back.
3. OIT flyPad: Settings > Realism shows the option.
4. `npx vitest run fbw-a380x/src/systems/instruments/src/MFD/FMC/PilotStoredElements.spec.ts` (17 tests), and the full `npm test` on Node 22+ (the two existing specs above load again).

DIRECT TO: ABEAM points, CRS IN and CRS OUT

The three DIRECT TO options of the A380 FCOM, on the shared flight plan code

Branch	feature/a380/mfd/direct-to-options
Aircraft	A380X (flight plan functions shared with the A32NX FMGC)
Commits	f98743703 feat(a380x/mfd): DIRECT TO with ABEAM points, CRS IN and CRS OUT
Base	pr/10-a380x-mfd-fcom-pages (first built on the old fork master 280800cf7)
Status	Merged in develop. 10 unit tests; harness-checked; not yet flown.

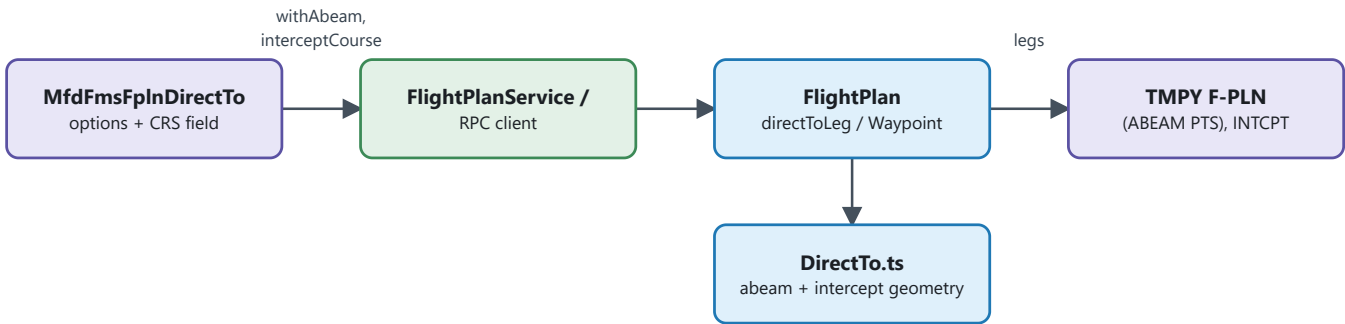
Purpose

The DIRECT TO page showed DIRECT WITH ABEAM, CRS IN and CRS OUT, but they could not be selected (no back end).

Specification (A380 FCOM DSC-22-FMS-20-30, DIRECT TO)

- DIRECT WITH ABEAM:** for each bypassed waypoint, an abeam point on the direct leg, named AB + 5 characters of the waypoint; the temporary F-PLN shows "(ABEAM PTS)".
- CRS IN / CRS OUT:** a course entered into / out of the target (format NNN or NNNT, FCOM defaults). The FMS computes an intercept point (INTCPT) on that course and flies direct to it, then along the course. No intercept point when the course cannot be intercepted (more than 160 deg, parallel or behind): the leg is a course to / from the target.
- The flight plan functions are shared (`fmgc/.../plans/DirectTo.ts`): `abeamPointOnLeg`, `directToInterceptPoint`, new leg factories and a pending-abeam flag cleared when the TMPY is inserted.

Component added or changed by the feature MSFS / simulator Aircraft system (systems host, FMS, FWS) Display / page
Data, LVar, storage Crew, flypad, external program



Architecture

A CI (course to intercept) leg was tried first: FBW's Geo.legIntercept picks a bogus start point (NaN). A computed INTCPT waypoint is used instead. Not done: the HDG-engaged / NAV-armed variant and the ADJUST DESIRED TRK OR HDG message; the intercept point is not recomputed when the track changes.

PR description (from the PR draft)

From `pr-drafts/14-direct-to-options.md`, section "Summary of Changes" onwards.

Summary of Changes

The DIRECT TO page showed the three options of the A380 FCOM (DSC-22-FMS-20-30, DIRECT TO) but they could not be selected.

- **DIRECT WITH ABEAM:** an abeam point on the direct leg for each bypassed waypoint, named AB + 5 characters of the waypoint; the temporary F-PLN shows "(ABEAM PTS)".
- **CRS IN / CRS OUT:** a course into / out of the target (entry format NNN or NNNT, FCOM default courses). The FMS computes an intercept point (INTCPT) on that course, flies direct to it and then along the course. When the course cannot be intercepted (more than 160 deg, parallel or behind), the leg is a course to / from the target without an intercept point.
- The flight plan functions are shared (`fmgc/.../plans/DirectTo.ts`: abeam point on a leg, intercept point), with 10 unit tests.

Cockpit API Changes

None.

Screenshots (if necessary)

TO ADD: the DIRECT TO page with each option and the resulting TMPY F-PLN and ND.

References

- A380 FCOM DSC-22-FMS-20-30, DIRECT TO page and its options.

Additional context

A CI leg was tried first: FBW's `Geo.legIntercept` returns a bogus start point (NaN) for it, so a computed intercept waypoint is used. Not done: the HDG-engaged / NAV-armed variant, the ADJUST DESIRED TRK OR HDG message, recomputing the intercept when the track changes.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. DIRECT TO a waypoint further along the plan WITH ABEAM: abeam points AB... in the TMPY, "(ABEAM PTS)".
2. CRS IN 090 to a waypoint ahead: INTCPT then the waypoint; CRS OUT: the waypoint then the course.
3. A course behind or parallel: no INTCPT, course to / from the target.

Fuel jettison

FQMS fuel jettison with the JTSN GW, the ECAM memo and procedures

Branch	feature/a380/fuel-jettison
Aircraft	A380X
Commits	b04a3851d feat(a380x): fuel jettison with JTSN GW, ECAM memo and procedures
Base	pr/10-a380x-mfd-fcom-pages (first built on the old fork master 280800cf7)
Status	Merged in develop. Node-simulated; deployed; in-sim test pending.

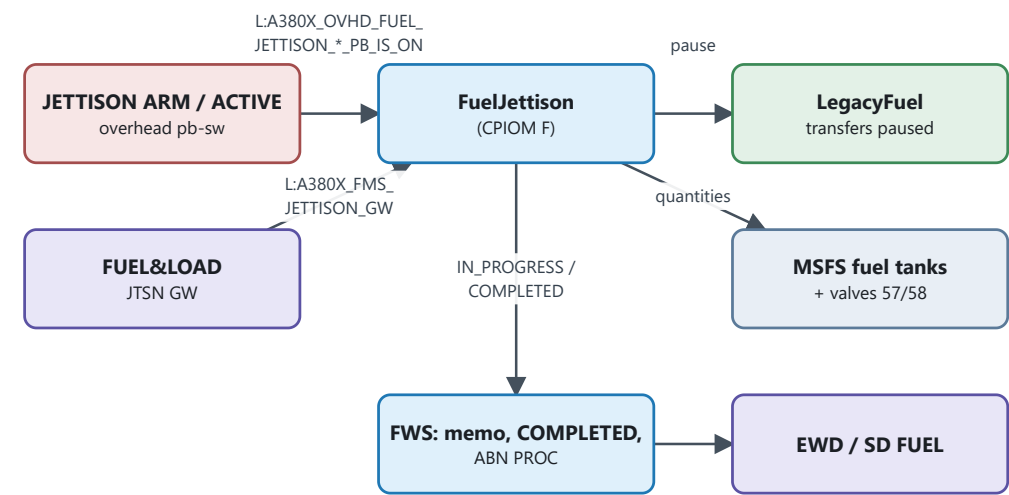
Purpose

FBW had the overhead JETTISON ARM / ACTIVE pushbuttons only (they toggled LVars). The JTSN GW field of FUEL&LOAD was disabled.

Specification (A380 FCOM DSC-28-40 Fuel jettison)

- CPIOM F `FuelJettison`: with ARM and ACTIVE on, fuel leaves the transfer tanks (proportionally) at the FCOM rate; the feed tanks are never jettisoned.
- Trim tank: held at the forward T.O/LDG CG limit (from the airframe data) and resumed at +0.5 %.
- Stops at the JTSN GW, when the transfer tanks are empty, or when a pb-sw is set OFF; COMPLETED latches until both pb-sw are OFF. The other fuel transfers pause during the jettison.
- The MSFS fuel system has no overboard outlet: the tank quantities are set directly (valves 57/58 open for the SD).
- FUEL&LOAD JTSN GW entry: from ZFW + 11 000 lb to the maximum GW (FCOM DSC-22-FMS-20-100), sent to the FQMS.
- ECAM: memo JETTISON IN PROGRESS; sensed FUEL JETTISON COMPLETED (ACTIVE OFF, ARM OFF, sensed); ABN PROC FUEL JETTISON (FUEL menu) with its FWS entry, the ARM / ACTIVE lines sensed. Cockpit: the ACTIVE ON light goes off when completed.

■ Component added or changed by the feature ■ MSFS / simulator ■ Aircraft system (systems host, FMS, FWS) ■ Display / page
■ Data, LVar, storage ■ Crew, flypad, external program



Architecture

Interfaces

LVar	Written by	Meaning
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L:A380X_FMS_JETTISON_GW	FMC	JTSN GW in kg
L:A380X_FUEL_JETTISON_IN_PROGRESS	FuelJettison	jettison running
L:A380X_FUEL_JETTISON_COMPLETED	FuelJettison	stopped by the FQMS, until both pb-sw are OFF

Limitations

- FAULT, VLV NOT CLOSED and INOP SYS are not wired (FBW has no jettison valve failures).
- The behaviour XML change (ACTIVE light) needs a full MSFS restart.

PR description (from the PR draft)

From `pr-drafts/15-fuel-jettison.md`, section "Summary of Changes" onwards.

Summary of Changes

FBW had the overhead JETTISON ARM / ACTIVE pb-sw only. Fuel jettison per the A380 FCOM DSC-28-40:

- **FQMS (CPIOM F FuelJettison)**: with ARM and ACTIVE on, fuel leaves the transfer tanks (proportionally) at the FCOM rate; the feed tanks are never jettisoned. The trim tank is held at the forward T.O/LDG CG limit and resumed at +0.5 %. The jettison stops at the JTSN GW, when the transfer tanks are empty or when a pb-sw is set OFF; COMPLETED latches until both are OFF. The other transfers pause meanwhile. The MSFS fuel system has no overboard outlet, so the tank quantities are set directly (valves 57 / 58 open for the SD).
- **FUEL&LOAD**: the JTSN GW field, from ZFW + 11 000 lb to the maximum GW (DSC-22-FMS-20-100), sent to the FQMS.
- **ECAM**: memo JETTISON IN PROGRESS; FUEL JETTISON COMPLETED (ACTIVE OFF, ARM OFF, sensed); ABN PROC FUEL JETTISON (FUEL menu) with its FWS entry, the ARM / ACTIVE lines sensed.
- **Cockpit**: the ACTIVE ON light goes off when the FQMS completes the jettison.

Cockpit API Changes

New LVars: `L:A380X_FMS_JETTISON_GW` (kg, FMC), `L:A380X_FUEL_JETTISON_IN_PROGRESS`, `L:A380X_FUEL_JETTISON_COMPLETED` (FQMS). Reads `L:A380X_OVHD_FUEL_JETTISON_ARM_PB_IS_ON` / `_ACTIVE_PB_IS_ON`.

Screenshots (if necessary)

TO ADD: FUEL&LOAD JTSN GW, the SD FUEL page during a jettison, the memo, the COMPLETED procedure, the ABN PROC procedure.

References

- A380 FCOM DSC-28-40 (fuel jettison), DSC-22-FMS-20-30 / -20-100 (FUEL&LOAD JTSN GW), PRO-ABN-ECAM-10-28 / -20-28 (procedures).

Additional context

FAULT, VLV NOT CLOSED and INOP SYS are not modelled (FBW has no jettison valve failures). The behaviour XML change needs a full MSFS restart.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. In flight, JTSN GW = GW - 20 t, ARM then ACTIVE: memo, fuel decreasing on the SD, trim tank held at the CG limit.
2. At the JTSN GW: jettison stops, COMPLETED procedure, ACTIVE light off; both pb-sw OFF clears it.
3. ABN PROC > FUEL > FUEL JETTISON: the procedure shows, ARM / ACTIVE lines tick with the pb-sw.

KCCU keyboard entries

MFD entry fields validated by ENT and cancelled by ESC, independent of the display focus

Branch	fix/a380/mfd/kccu-keyboard-entry
Aircraft	A380X (MFD, OIT, ND OANS panel)
Commits	b45746fa7 fix(a380x/mfd): KCCU keyboard entries validated with ENT, ESC cancels the edition
Base	pr/10-a380x-mfd-fcom-pages (first built on the old fork master 280800cf7)
Status	Merged in develop. Harness-tested (8 scenarios); in-sim confirmation pending.

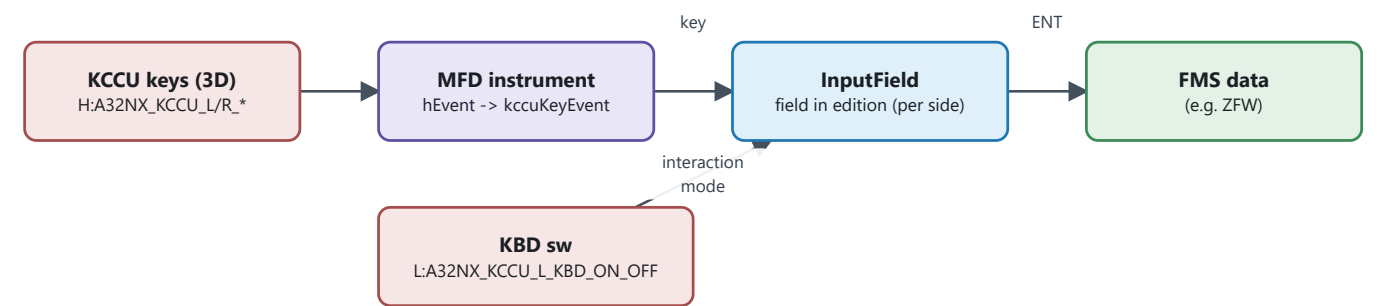
Problem

In the sim, a value typed on the KCCU keys (e.g. a new ZFW) was not taken. An entry was only validated by the blur event of the display, and a click on a 3D KCCU key can end the edition before the key reaches the field.

Specification (A380 FCOM DSC-31-30-20 KCCU)

- ENT validates the entry directly (no blur event needed).
- ESC "cancels the current field edition. The field returns to the last valid value" (FCOM P 8). It entered the old value again before (running the entry handler).
- With the KBD sw ON ("The KCCU keyboard is active"), a blur of the display does not end the edition: ENT, ESC or the selection of another field end it (the previous field is validated). One field in edition per side.
- A click on a field selects it even when the element already has the focus.
- With the KBD sw OFF, the PC keyboard works as before (typing + Enter, or a click elsewhere validates).

■ Component added or changed by the feature ■ MSFS / simulator ■ Aircraft system (systems host, FMS, FWS) ■ Display / page
■ Data, LVar, storage ■ Crew, flypad, external program



Architecture

PR description (from the PR draft)

From `pr-drafts/16-kccu-keyboard-entry.md`, section "Summary of Changes" onwards.

Summary of Changes

A value typed on the KCCU keys was not entered: an entry field was only validated by the blur event of the display, and in the sim a click on a 3D KCCU key can end the edition before the key reaches the field. `InputField` now follows the A380 FCOM (DSC-31-30-20, KCCU):

- ENT validates the entry directly.
- ESC "cancels the current field edition. The field returns to the last valid value" (it entered the old value again before).
- With the KBD sw ON ("The KCCU keyboard is active"), a blur of the display does not end the edition: ENT, ESC or the selection of another field end it (the previous field is then validated); one field in edition per side.
- A click on a field selects it even when the element already has the focus.
- KBD sw OFF: the PC keyboard works as before.

Cockpit API Changes

None.

References

- A380 FCOM DSC-31-30-20 (KCCU keyboard, ON/OFF sw, ENT, ESC).

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. KBD sw ON: click the FUEL&LOAD ZFW field, type 3 3 0 on the KCCU, ENT: ZFW 330.0.
2. Type 1 2 then ESC: the previous value stays, nothing is entered.
3. Type in ZFW, click ZFWCG, type, ENT: both entered.
4. KBD sw OFF: PC keyboard typing + Enter works.

ECAM procedures from the A380 FCOM

ABN PROC (not-sensed) procedures and the sensed alerts that have FCOM action lines

Branch	feature/a380/fws/abn-proc-procedures
Aircraft	A380X
Commits	00b3a980d feat(a380x/fws): ECAM procedures from the A380 FCOM for the ABN PROC and the sensed alerts
Base	pr/10-a380x-mfd-fcom-pages (first built on the old fork master 280800cf7)
Status	Merged in develop. Built, tsc/lint clean, consistency-checked; deployed, in-sim test pending.

Problem

Thirteen ABN PROC procedures were marked WIP with no lines, and a not-sensed procedure is only shown on the EWD if it has an entry in `FwsAbnormalNonSensed`: activating them showed nothing. 102 alerts wired in the FWS had no items.

Specification

ABN PROC (A380 FCOM PRO-ABN-ECAM-20; texts from PRO-ABN-ECAM-10)

Procedure	Menu	Notes
MISC DITCHING	first page	FCOM text incl. approach, 2000 / 500 ft, after ditching
MISC FORCED LANDING	first page	engines running case
MISC CKPT WINDOW CRACKED / ELEC ARCING	MISCELLANEOUS	external / internal layer, AICU 1 / 2
MISC OIS FAULT	MISCELLANEOUS	OIT / laptop restart
FUEL LEAK	FUEL	the FUEL LEAK DETECTED procedure; LAND ANSA
FUEL MAN BALANCING PROCEDURE	FUEL	DELAY T.O on ground; feed / inner / mid tank balancing in flight
L/G LDG WITH ABNORM ... (5 procedures)	L/G	WHEEL SD page
NAV IR ALIGNMENT IN ATT MODE	NAV	
ENG HI VIBRATIONS (new)	ENG	in the FCOM ENG list, was missing

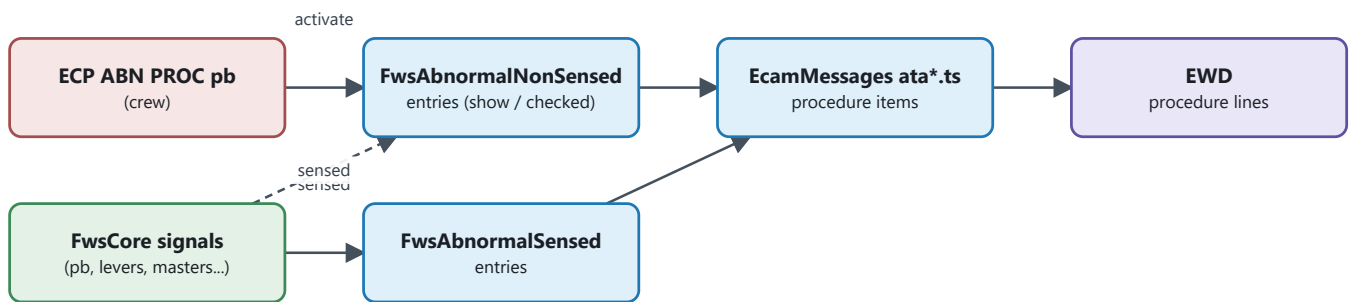
Sensed alerts (A380 FCOM PRO-ABN-ECAM-10)

- Of the 102 wired alerts without items, about 85 are "Crew awareness" in the FCOM (title only, plus STATUS / redundancy lines): their empty item list is correct. 4 are not in the 2011 FCOM (HOT AIR 1(2) OFF / VLV OPEN, T.O V1/VR/V2 DISAGREE, OVERSPEED LOAD ANALYSIS REQUIRED) and stay empty.
- Filled: COND HOT AIR 1(2) FAULT, COND TEMP CTL FAULT, ENG 1(2)(3)(4) FAIL (in flight / on ground, engine number in the lines, sensed thrust lever / master / fire pb / agent), ENG ALL ENGINES FAILURE (FCOM ALL ENG FLAME OUT, fuel remaining / no fuel, forced landing and ditching; nesting capped at three levels for the EWD width).

FCOM conventions applied

- Upper case "IF ...:" is an ECAM condition line; lower case "If ...:" is a display condition (it selects which lines are shown, in `whichItemsToShow`).
- Headers like "AT 2000 FT AGL :-" are centred sub-headlines; lines are at most 41 characters (EWD width).
- A line is sensed only where the FWS has the signal (seat belts, masters, fire pb, agents, park brake, RAM AIR, L/G lever, ground spoilers, GPWS / TERR off, A/THR, thrust levers, APU master / bleed, EVAC, batteries).

■ Component added or changed by the feature
 ■ MSFS / simulator
 ■ Aircraft system (systems host, FMS, FWS)
 ■ Display / page
■ Data, LVar, storage
■ Crew, flypad, external program



Architecture

Verification

- Every item list vs its show / checked arrays checked by script over all 611 arrays of the two FWS files: consistent (one harmless pre-existing extra in VOLCANIC ASH).
- systems host, EWD, PFD, SD build; tsc systems-host clean; ESLint and Prettier clean.

Limitations

- Deferred procedures, limitations and STATUS of the new procedures are not written yet.
- FLUCTUATING VERTICAL SPEED stays WIP (not in the 2011 FCOM).
- Where the FCOM lets the ECAM pick a side (e.g. L/G affected side) and FBW cannot sense it, a condition line is used.

PR description (from the PR draft)

From `pr-drafts/17-ecam-procedures.md`, section "Summary of Changes" onwards.

Summary of Changes

A not-sensed procedure is only displayed with its entry in `FwsAbnormalNonSensed`: the ABN PROC procedures marked WIP had neither lines nor entry and showed nothing when activated. The procedures now carry the A380 FCOM text and their FWS entries (sensed lines where the FWS has the signal).

ABN PROC (FCOM PRO-ABN-ECAM-20; texts PRO-ABN-ECAM-10): MISC DITCHING, FORCED LANDING, CKPT WINDOW CRACKED, CKPT WINDOW ELEC ARCING, OIS FAULT; FUEL LEAK (the FUEL LEAK DETECTED procedure, LAND ANSA), MAN BALANCING PROCEDURE; L/G LDG WITH ABNORM NOSE L/G, WITH 2 ABNORM L/Gs ON SAME SIDE, WITH 1 ABNORM WING OR BODY L/G, WITH 2 ABNORM BODY L/Gs, WITH 2 ABNORM WING L/Gs (WHEEL page); NAV IR ALIGNMENT IN ATT MODE; new ENG HI VIBRATIONS (in the FCOM ENG list, was missing).

Sensed alerts (FCOM PRO-ABN-ECAM-10): of the 102 wired alerts without lines, about 85 are "Crew awareness" in the FCOM (title only): left empty. Filled: COND HOT AIR 1(2) FAULT, COND TEMP CTL FAULT, ENG 1(2)(3)(4) FAIL (in flight / on ground, the engine number in the lines, thrust lever / master / fire pb / agent sensed, L/G UP after lift-off), ENG ALL ENGINES FAILURE (the FCOM ALL ENG FLAME OUT: fuel remaining / no fuel, relight, forced landing and ditching).

FCOM conventions: upper case "IF ...:" lines are ECAM condition lines; lower case "If ...:" only select the lines shown. Headers ("AT 2000 FT AGL :") are centred sub-headlines; lines fit the 41-character EWD width (nesting capped at three levels).

Cockpit API Changes

None.

Screenshots (if necessary)

TO ADD: each ABN PROC procedure on the EWD, ENG 1 FAIL in flight and on ground, ALL ENGINES FAILURE.

References

- A380 FCOM PRO-ABN-ECAM-10 (procedures), PRO-ABN-ECAM-20 (not-sensed ABN PROC list).

Additional context

Not done: the deferred procedures, limitations and STATUS of the new procedures; FLUCTUATING VERTICAL SPEED stays WIP (not in the 2011 FCOM). HOT AIR 1(2) OFF / VLV OPEN, T.O V1/VR/V2 DISAGREE and OVERSPEED LOAD ANALYSIS REQUIRED are not in this FCOM and stay without lines.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. ECP ABN PROC: open each procedure of the list, activate it: the lines show; sensed lines tick (e.g. SIGNS, ALL ENG MASTERS, PARK BRK).
2. Fail an engine in flight: ENG x FAIL with AUTO RELIGHT IN PROGRESS, THR LEVER x / ENG x MASTER; on ground: the ground lines.
3. Fail all engines in flight: ALL ENGINES FAILURE, LAND ASAP, the full procedure scrolls.
4. HOT AIR 1 pb OFF: COND HOT AIR 1 FAULT with HOT AIR 1 ON.

A32NX RAAS runway awareness callouts

The Honeywell RAAS option of the EGPWS, with the runways of the sim and a generated voice

Branch	pr/18-a32nx-raas
Aircraft	A32NX (A380X OIT flyPad: settings option only)
Commits	b7a7cd707 feat(a32nx/egpws): add the RAAS runway awareness callouts; 22d963d5c fix(a380x/oit): add the RAAS pin program option to the OIT settings pages
Base	upstream master 2baa2b35e (independent)
Status	Voice confirmed in the sim; CI checks pass. Before opening: confirm the licence of the Piper voice model.

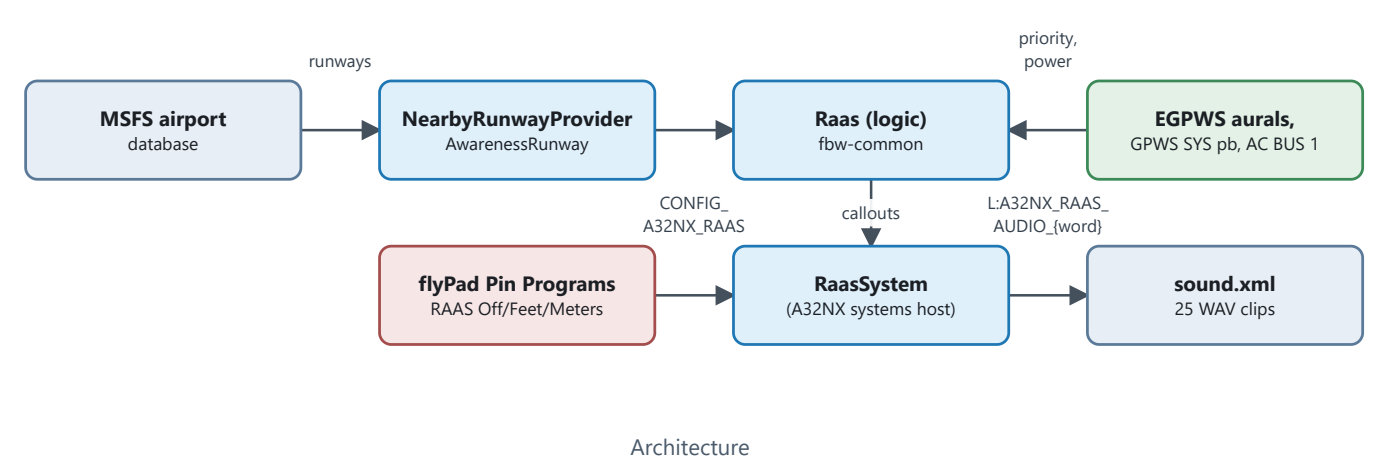
Purpose

The Runway Awareness and Advisory System is an option of the Honeywell EGPWS of the A320 (A320 FCOM GEN, aircraft configuration "RAAS"; used as a line-up aid in PRO-NOR-SOP-11 TAKEOFF RUNWAY CONFIRM). The A32NX did not have it.

Specification (Honeywell RAAS product description 060-4564-001, section 4)

- Approaching runway on the ground and in the air (750 to 300 ft, with the Airbus 450-350 ft suppress window).
- On runway; on taxiway above 40 kt; extended holding on runway.
- Distance remaining at landing and at a rejected takeoff; runway end.
- Approaching short runway and insufficient runway length, with their cautions; taxiway landing.
- The calls yield to the EGPWS aural and run while the EGPWC (AC BUS 1) is powered and the GPWS SYS pushbutton is not OFF.

Component added or changed by the feature MSFS / simulator Aircraft system (systems host, FMS, FWS) Display / page Data, LVar, storage Crew, flypad, external program



Voice

The callouts are played word by word: 25 WAV clips, through the predefined `custom*_sound_NN` event slots of the Wave Data pipeline. The clips are generated by `tools/raas-voice/generate.py` with the Piper text-to-speech engine and its `en_US-ljspeech-medium` voice, trained on the public-domain LJ Speech dataset.

Interfaces

Name	Type	Meaning
------	------	---------

<code>L:A32NX_RAAS_AUDIO_{word}</code>	LVar (Bool)	true while the RAAS says that word (a320-simvars.md, GPWS / TAWS)
<code>CONFIG_A32NX_RAAS</code>	flyPad setting	OFF / FEET (default) / METERS: the unit of the distance callouts

Tests

16 unit tests: `fbw-common/src/systems/shared/src/runway-awareness/Raas.spec.ts`.

Limitations

- Open before the PR: the licence of the Piper voice model `en_US-1jspeech-medium` (its model card); the WAV clips must be redistributable under the repository licence.

PR description (from the PR draft)

From `pr-drafts/18-a32nx-raas.md`, section "Summary of Changes" onwards.

Summary of Changes

The Runway Awareness and Advisory System is an option of the A320's Honeywell EGPWS (A320 FCOM GEN aircraft configuration "RAAS"; used as a line-up aid in PRO-NOR-SOP-11 TAKEOFF RUNWAY CONFIRM). This adds it to the A32NX, implemented from the Honeywell RAAS product description 060-4564-001, section 4:

- **Callouts:** approaching runway on the ground and in the air (750 to 300 ft, with the Airbus 450-350 ft suppress window), on runway, on taxiway above 40 kt, extended holding on runway, distance remaining at landing and at a rejected takeoff, runway end, approaching short runway and insufficient runway length with their cautions, taxiway landing.
- **Runways:** from the sim's own airport database, so no airport map (Navigraph AMDB) is needed.
- **Priorities and power:** the calls yield to the EGPWS aural and run while the EGPWS (AC BUS 1) is powered and the GPWS SYS pushbutton is not OFF.
- **Voice:** played word by word (25 WAV clips) through the predefined `custom*_sound_NN` event slots of the Wave Data pipeline. The clips are generated by `tools/raas-voice/generate.py` with the Piper text-to-speech engine and its `en_US-ljspeech-medium` voice, trained on the public-domain LJ Speech dataset.
- **Setting:** flyPad Aircraft Options / Pin Programs, RAAS: Off, Feet (default) or Meters (the unit of the distance callouts).
- **OIT:** the A380X OIT flyPad gets the new pin program option too (off, as on the A380X EFB); without it the FBW_TYPECHECK build fails.
- 16 unit tests (`fbw-common/src/systems/shared/src/runway-awareness/Raas.spec.ts`).

Cockpit API Changes

- New LVars `L:A32NX_RAAS_AUDIO_{word}` (Bool, true while the RAAS says that word), documented in `fbw-a32nx/docs/a320-simvars.md` (GPWS / TAWS).
- New flyPad setting `CONFIG_A32NX_RAAS` (OFF / FEET / METERS).

Screenshots (if necessary)

TO ADD: the flyPad Pin Programs RAAS option; a short video or audio of "approaching runway", "on runway" and a distance remaining callout.

References

- Honeywell Runway Awareness and Advisory System (RAAS) product description 060-4564-001, section 4.
- A320 FCOM GEN (aircraft configuration: RAAS), PRO-NOR-SOP-11 (TAKEOFF RUNWAY CONFIRM).

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. flyPad Aircraft Options / Pin Programs: RAAS Feet.
2. Taxi towards a runway: "approaching runway ..." at the holding point; line up: "on runway ...".
3. Taxi faster than 40 kt on a taxiway: the taxiway caution.
4. Land: the distance remaining callouts during the rollout (in feet, then with Meters in meters).
5. GPWS SYS pushbutton OFF, or RAAS Off: no callouts.
6. `npx vitest run fbw-common/src/systems/shared/src/runway-awareness/Raas.spec.ts` (16 tests).

A380X OANS RWY AHEAD without an airport map

The RWY AHEAD advisory from the runways of the sim when no Navigraph airport map is loaded

Branch	pr/19-a380x-oans-rwy-ahead-without-map
Aircraft	A380X
Commits	aac868b24 feat(a380x/oans): show RWY AHEAD without an airport map
Base	pr/18-a32nx-raas (uses its NearbyRunwayProvider and AwarenessRunway)
Status	CI checks pass. Opens after PR 18 (it waits for its voice model licence check too).

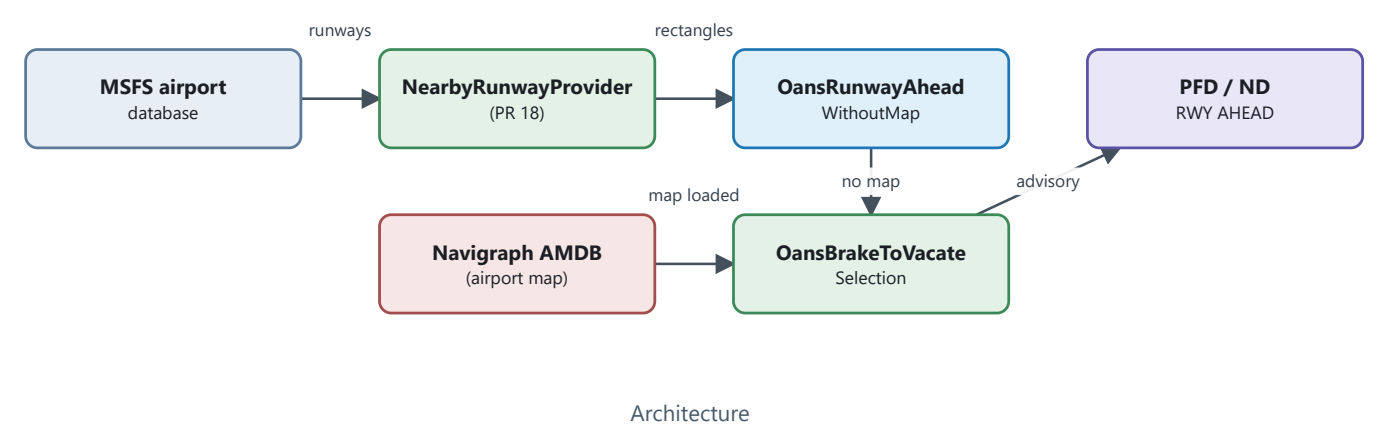
Purpose

The OANS RWY AHEAD advisory needs the Navigraph airport map (AMDB) and stays silent without it, so most users never get it.

Specification

- A380 FCOM DSC-31-20-30-100: ND message RWY AHEAD : CHANGE MODE. This FCOM edition does not describe the detection itself: the rule is the existing FBW OANS RWY AHEAD rule.
- While no airport map is loaded, OANS runs that rule against the runway rectangles of the sim airport database: the volume from the nose to 7 s ahead, 60 m wide, from 1 to 40 kt on the ground, 30 s at most.
- The advisory is issued on the PFD and the ND as with the map. With an airport map loaded, nothing changes.

Component added or changed by the feature MSFS / simulator Aircraft system (systems host, FMS, FWS) Display / page
Data, LVar, storage Crew, flypad, external program



Tests

4 unit tests: `fbw-common/src/systems/instruments/src/OANC/OansRunwayAheadWithoutMap.spec.ts`.

PR description (from the PR draft)

From `pr-drafts/19-a380x-oans-rwy-ahead-without-map.md`, section "Summary of Changes" onwards.

Summary of Changes

The OANS RWY AHEAD advisory needs the Navigraph airport map (AMDB) and stays silent without it, so most users never get it.

While no airport map is loaded, OANS now runs the same rule as with the map (the volume from the nose to 7 s ahead, 60 m wide, from 1 to 40 kt on the ground, 30 s at most) against the runway rectangles of the sim's own airport database, through the shared `NearbyRunwayProvider` (added by #[PR 18 number], the RAAS PR), and issues the advisory on the PFD and ND the same way. With an airport map loaded, nothing changes.

4 unit tests (`fbw-common/src/systems/instruments/src/OANC/OansRunwayAheadWithoutMap.spec.ts`).

Cockpit API Changes

None.

Screenshots (if necessary)

TO ADD: RWY AHEAD on the PFD and ND while taxiing onto a runway without an airport map loaded.

References

- A380 FCOM DSC-31-20-30-100 (ND message RWY AHEAD : CHANGE MODE). This FCOM edition does not describe the detection itself: the rule is FBW's existing OANS RWY AHEAD rule, applied to the sim's runways.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. No Navigraph airport map loaded (no subscription, or OANS map not loaded).
2. Taxi slowly towards a runway: RWY AHEAD on the PFD and ND when the runway is within 7 s ahead.
3. Above 40 kt, or airborne: no advisory. With the airport map loaded: the existing behaviour.
4. `npx vitest run fbw-common/src/systems/instruments/src/OANC/OansRunwayAheadWithoutMap.spec.ts` (4 tests).

A380X computed ALTN fuel and time

The FCOM default computation of the fuel and time to the alternate, instead of a fixed 6.5 t

Branch	pr/20-a380x-altn-fuel-computation
Aircraft	A380X (MFD FUEL&LOAD and ALTERNATE pages)
Commits	3c3ef82b7 feat(a380x/mfd): computed ALTN fuel and time (FCOM default computation)
Base	pr/10-a380x-mfd-fcom-pages (FUEL&LOAD page)
Status	CI checks pass.

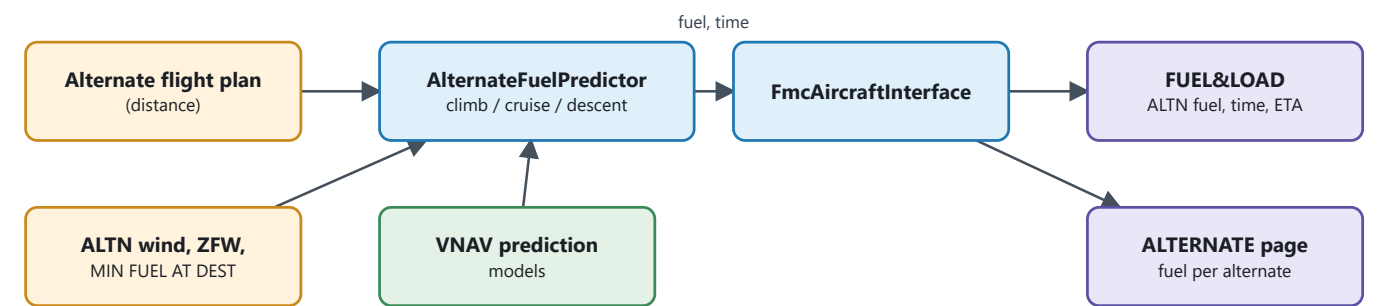
Purpose

The ALTN fuel of the FUEL&LOAD page was a fixed 6.5 t, whatever the alternate.

Specification (A380 FCOM DSC-22-FMS-20-30, FUEL&LOAD page)

- ALTN fuel (time): the trip from the primary destination to the landing at the alternate, computed with CI 0.
- Cruise at FL220 when the alternate flight plan distance is less than 200 NM, FL310 otherwise; lowered when the distance is too short for the climb and the descent.
- Climb at climb thrust, cruise, idle descent, with the VNAV prediction models of the aircraft at the FMS CI 0 speeds (290 kt, 288 kt in descent, M0.84, 250 kt below FL100).
- Distance: the alternate flight plan without its missed approach; the direct distance without an alternate route. Wind: the ALTN wind. Weight: at destination (ZFW + MIN FUEL AT DEST).
- No ALTN fuel without a ZFW; recomputed only when its inputs change.

■ Component added or changed by the feature ■ MSFS / simulator ■ Aircraft system (systems host, FMS, FWS) ■ Display / page
■ Data, LVar, storage ■ Crew, flypad, external program



Architecture

Displays

- FUEL&LOAD: the ALTN time next to the ALTN fuel, and the arrival at the alternate (destination + ALTN time).
- ALTERNATE page: the fuel to each alternate, with the same computation.

Tests

5 unit tests: `fbw-a380x/src/systems/instruments/src/MFD/FMC/AlternateFuelPredictor.spec.ts`.

PR description (from the PR draft)

From `pr-drafts/20-a380x-altn-fuel-computation.md`, section "Summary of Changes" onwards.

Summary of Changes

The ALTN fuel of the FUEL&LOAD page was a fixed 6.5 t. The A380 FCOM (DSC-22-FMS-20-30, FUEL&LOAD page) defines it: the fuel (time) for the trip between the primary destination and the landing at the alternate, computed with CI 0, at FL220 when the alternate flight plan distance is less than 200 NM, FL310 otherwise.

- **AlternateFuelPredictor:** climb at climb thrust, cruise and idle descent with the aircraft's VNAV prediction models, at the FMS CI 0 speeds (290 kt, 288 kt in descent, M0.84, 250 kt below FL100). The cruise level is lowered when the distance is too short for the climb and the descent.
- **FMS:** over the alternate flight plan distance (without its missed approach; the direct distance when there is no alternate route), with the ALTN wind and the weight at destination (ZFW + MIN FUEL AT DEST). No default ALTN without a ZFW; recomputed only when its inputs change.
- **FUEL&LOAD:** the ALTN time next to the ALTN fuel, and the arrival at the alternate (destination + ALTN time).
- **ALTERNATE page:** the fuel to each alternate, with the same computation.
- 5 unit tests (`AlternateFuelPredictor.spec.ts`).

Cockpit API Changes

None.

Screenshots (if necessary)

TO ADD: FUEL&LOAD with the computed ALTN fuel and time and the alternate ETA; the ALTERNATE page with the fuel of each alternate.

References

- A380 FCOM DSC-22-FMS-20-30 (FUEL&LOAD page: ALTN fuel and time, default computation at CI 0, FL220 below 200 NM, FL310 above).

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. Flight plan with an alternate, ZFW entered: FUEL&LOAD shows a computed ALTN fuel and time (no longer 6.5 t) and the ALTN ETA.
2. An alternate closer than 200 NM against one further away: the lower fuel; the ALTERNATE page lists the fuel of each alternate.
3. No ZFW: no ALTN fuel.
4. `npx vitest run fbw-a380x/src/systems/instruments/src/MFD/FMC/AlternateFuelPredictor.spec.ts` (5 tests).

A380X company datalink reply time

REQUEST PENDING and NO COMPANY REPLY for the company F-PLN, wind and T.O data requests

Branch	pr/21-a380x-company-datalink-delay
Aircraft	A380X (MFD, flyPad); A32NX MCDU unchanged
Commits	3cc600fd6 feat(a380x): company datalink reply time for the F-PLN, wind and T.O data requests; e17ebd25b fix(a380x/mfd): fix the tsc-strict errors in the FMC; 199d44f3b fix(a380x/oit): add the company datalink reply time option to the OIT settings pages
Base	pr/12-a320-takeoff-calculator (company T.O data link; PR 10 below it: CPNY F-PLN and WIND requests)
Status	CI checks pass.

Purpose

The A380 FCOM (DSC-22-FMS-10-40-90) describes the company requests as datalink messages: the button shows REQUEST PENDING... until the answer, and NO COMPANY REPLY appears without an answer within 4 min. The FMS answered them instantly.

Specification

- flyPad Settings > Realism > Company Datalink Reply Time (A380X): Instant / Fast / Real, like the ADIRS align time.
- CPNY F-PLN request: REQUEST PENDING... from the click, one request at a time, the SimBrief OFP delivered after the reply time, NO COMPANY REPLY at 4 min (a failed download used to fail silently).
- CPNY WIND request: the answer after the reply time.
- T.O data: the flyPad answer reaches the FMS not before the reply time of SEND T.O REQUEST (only the transit time without a request). The A32NX MCDU uplink stays instant.

Setting	Reply after a request	Transit without a request
Instant	0	0
Fast	5-15 s	1-3 s
Real (default)	60-120 s	5-15 s

Airbus gives no figure: these are typical airline values, below the 4 min limit of the FCOM (`CompanyDatalinkDelay` in fbw-sdk).

- Component added or changed by the feature

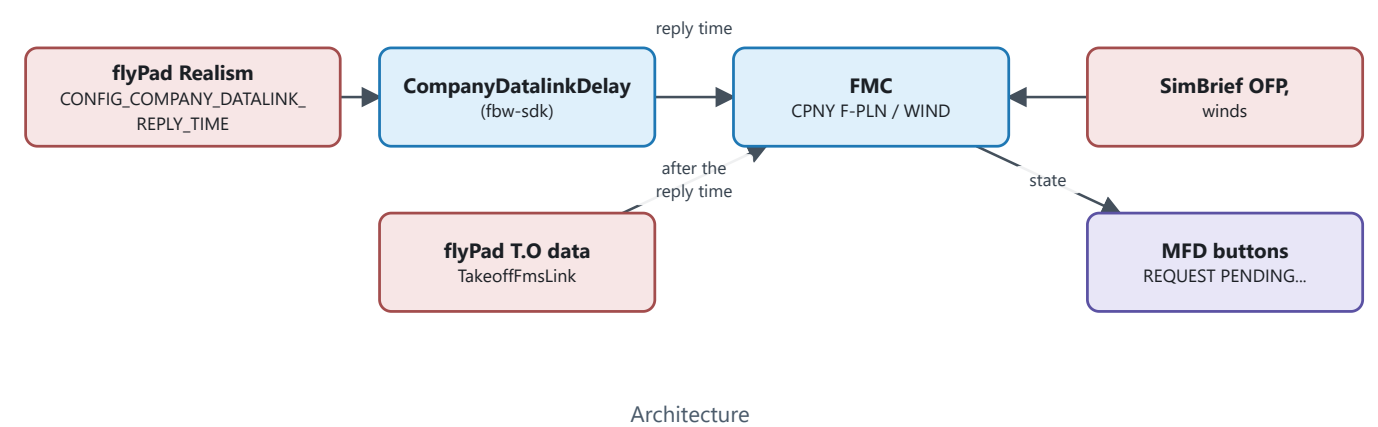
MSFS / simulator

Aircraft system (systems host, FMS, FWS)

Display / page

Data, LVar, storage

Crew, flypad, external program



Interfaces

Name	Type	Meaning
CONFIG_COMPANY_DATAINK_REPLY_TIME	flyPad setting	INSTANT / FAST / REAL

The OIT flyPad (OITlegacy) gets the new realism option too, with the A380X EFB value.

PR description (from the PR draft)

From `pr-drafts/21-a380x-company-datalink-delay.md`, section "Summary of Changes" onwards.

Summary of Changes

The A380 FCOM (DSC-22-FMS-10-40-90) describes the company requests as datalink messages: the button shows REQUEST PENDING... until the answer, and NO COMPANY REPLY appears when there is no answer within 4 min. The FMS answered them instantly.

- **flyPad Settings > Realism > Company Datalink Reply Time (A380X):** Instant / Fast / Real, like the ADIRS align time. Reply time after a request: 0 / 5-15 s / 60-120 s (default Real); transit of a message sent without a request: 0 / 1-3 s / 5-15 s. Airbus gives no figure: these are typical airline values, below the 4 min limit (`CompanyDataLinkDelay` in `fbw-sdk`).
- **CPNY F-PLN request:** REQUEST PENDING... from the click, one request at a time, the SimBrief OFP delivered after the reply time, NO COMPANY REPLY at 4 min (a failed download used to fail silently).
- **CPNY WIND request:** the answer is delivered after the reply time.
- **T.O data:** the flyPad answer reaches the FMS not before the reply time of SEND T.O REQUEST (only the transit time without a request). As in PR 12, only the thrust selected for the takeoff run is sent. The A32NX MCDU uplink stays instant.
- The FMC `navigraphUsername` argument is typed `string | undefined` (`FBW_TYPECHECK tsc-strict`).
- The OIT flyPad (OITlegacy) gets the new realism option too, with the A380X EFB value; without it the `FBW_TYPECHECK` build fails.

Cockpit API Changes

None. New flyPad setting `CONFIG_COMPANY_DATA_LINK_REPLY_TIME (INSTANT / FAST / REAL)`.

Screenshots (if necessary)

TO ADD: the Realism setting; the CPNY F-PLN button showing REQUEST PENDING...; NO COMPANY REPLY after 4 min without an answer.

References

- A380 FCOM DSC-22-FMS-10-40-90 (company requests: REQUEST PENDING..., NO COMPANY REPLY after 4 min).

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. Setting Real: CPNY F-PLN REQUEST shows REQUEST PENDING... and the flight plan arrives after 1-2 min; Instant: at once.
2. SimBrief not reachable: NO COMPANY REPLY at 4 min.
3. CPNY WIND REQUEST: the winds arrive after the reply time.
4. SEND T.O REQUEST on the MFD, then SEND TO FMS on the flyPad: the data reach the FMS not before the reply time.
5. A32NX: the MCDU uplink takeoff data are still instant.

A380X MFD ATC COM pages and the SD mailbox

Every ATC COM page of the A380 FCOM on the FBW ATSU, and the SD mailbox to answer the ATC

Branch	pr/22-a380x-mfd-atccom
Aircraft	A380X (MFD, SD)
Commits	b3a72399e feat(a380x/mfd): ATC COM pages and the SD mailbox per the A380 FCOM; c-f66b105f fix(a380x/mfd): lint and strict types of the ATC COM pages and the SD mailbox; 703024a90 docs(changelog): add the ATC COM pages and the SD mailbox; fixes of the in-sim test: 899b8c6eb frame delete, label layout, ATIS auto update; c0326bbd0 texts sized for the FBW display font; 625a792d5 ATIS time from the ATIS message
Base	pr/10-a380x-mfd-fcom-pages (MFD FCOM layout and common components)
Status	CI checks pass. Tested in the sim on 2026-09-27: ATIS pages and the REQUEST / REPORT / EMERGENCY frames; logon, CPDLC with Hoppie and the SD mailbox answers not tested yet.

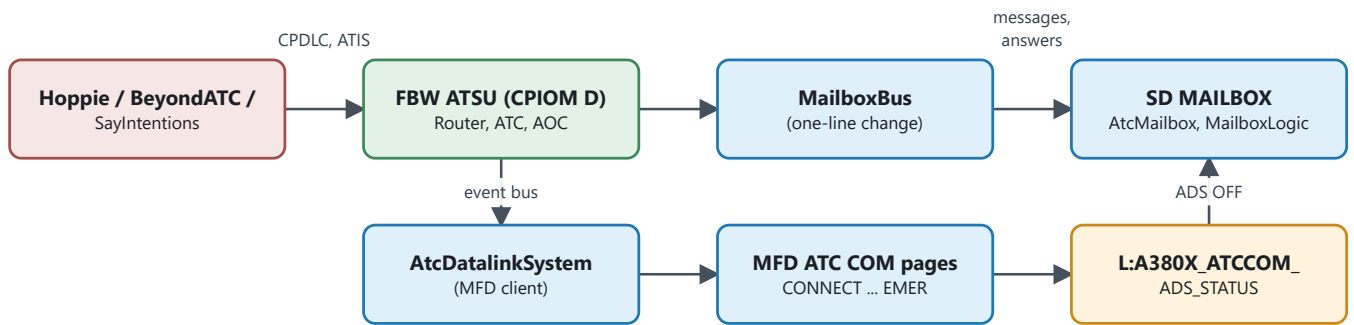
Purpose

The A380X MFD had placeholder ATC COM pages (connect, D-ATIS, message record) and the SD mailbox could not answer the ATC. The FBW ATSU (Hoppie, BeyondATC, SayIntentions through the flyPad ATSU/AOC settings) was already hosted by the A380X systems host.

Specification (A380 FCOM DSC-46-10-20-30 ATC COM pages, DSC-46-10-10-70 mailbox, DSC-46-10-20-40 messages)

Page	Content
Header	CONNECT / REQUEST / REPORT & MODIFY / MSG RECORD / ATIS / EMER, the ATC COM system menu
CONNECT	NOTIFICATION (logon: NOTIFYING, NOTIFICATION FAILED, the centers notified), CONNECTION STATUS (with the ADS status), MAX UPLINK DELAY
ATIS	LIST with three request areas (departure, arrival, alternate, following the FMS airports), AUTO UPDATE, UPDATE ALL, ATIS RECEIVED
MSG RECORD	LIST and ZOOM, ERASE ALL of the closed dialogues, MSG RECORD USED OFFSIDE
REQUEST	every FCOM frame mapped to its CPDLC downlink element, the DEPARTURE (DCL) and OCEANIC (OCL) clearances, XFR TO MAILBOX
REPORT & MODIFY	AUTO and MANUAL POSITION reports filled by the FMS, OTHER REPORTS, MODIFY of a prepared reply
EMERGENCY	MAYDAY, PANPAN, CANCEL EMER with the emergency frames; MAYDAY switches the ADS to EMERGENCY
SD MAILBOX	answer buttons per expected response (WILCO / STANDBY / UNABLE, AFFIRM / NEGATIVE, ROGER), SEND / CANCEL / CLOSE, DUE TO and FREETEXT replies, ACK / REFUSE for DCL and OCL, the FCOM status colours

 Component added or changed by the feature  MSFS / simulator  Aircraft system (systems host, FMS, FWS)  Display / page
 Data, LVar, storage  Crew, flypad, external program



Architecture

Interfaces

LVar	Written by	Meaning
L:A380X_ATCCOM_ADS_STATUS	MFD	0 ARMED, 1 CONNECTED, 2 OFF; displayed by the SD mailbox
L:A380X_MFD_{side}_ATCCOM_MSG_RECORD_ZOOM	MFD	an MFD shows a recorded message: MSG RECORD USED OFFSIDE on the other one

Documented in `fbw-a380x/docs/a380x-private-local-vars.md` (46 - Information Systems). `fbw-common` MailboxBus: `downlinkTransmit` on an uplink sends its existing open or failed response.

Tests

33 unit tests: request frames, entry formats, ATIS text and time, message record, reports, mailbox logic.

Limitations

- No ADS contract through the ATSU: the ADS status is local (ARMED / OFF, EMERGENCY).
- Parked: LOAD to SEC3 of a route clearance (the ATSU has no route loading) and the KCCU MAILBOX key.
- The PRINT functions show PRINTER NOT AVAIL in this branch; the ATIS printing is on the feature branch and needs the pedestal printer, which has no PR yet.

PR description (from the PR draft)

From `pr-drafts/22-a380x-mfd-atccom.md`, section "Summary of Changes" onwards.

Summary of Changes

Every ATC COM page of the A380 FCOM (DSC-46-10-20-30) is laid out on its FCOM figure and wired to the FBW ATSU, and the SD MAILBOX (DSC-46-10-10-70) becomes a client of the ATSU mailbox protocol (the same one as the A32NX DCU).

- **Header:** CONNECT / REQUEST / REPORT & MODIFY / MSG RECORD / ATIS / EMER, and the ATC COM system menu.
- **CONNECT:** NOTIFICATION (logon; NOTIFYING, NOTIFICATION FAILED; the centers notified), CONNECTION STATUS (with the ADS status), MAX UPLINK DELAY.
- **ATIS:** LIST, with three request areas (departure, arrival, alternate, following the FMS airports), AUTO UPDATE and UPDATE ALL, and ATIS RECEIVED.
- **MSG RECORD:** LIST and ZOOM, ERASE ALL of the closed dialogues, MSG RECORD USED OFFSIDE.
- **REQUEST:** every FCOM frame mapped to its CPDLC downlink element (climb and descent, altitude and block, direct, offset, weather deviation, heading and track, SID/STAR, speed, when can we expect, free text, voice contact, own separation, VMC descent, additional text), the DEPARTURE (DCL) and OCEANIC (OCL) clearances, XFR TO MAILBOX.
- **REPORT & MODIFY:** AUTO and MANUAL POSITION reports filled by the FMS, OTHER REPORTS, MODIFY of a prepared reply.
- **EMERGENCY:** MAYDAY, PANPAN and CANCEL EMER with the emergency frames; MAYDAY switches the ADS to EMERGENCY.
- **SD MAILBOX:** the answer buttons per expected response (WILCO / STANDBY / UNABLE, AFFIRM / NEGATIVE, ROGER), SEND / CANCEL / CLOSE, DUE TO and FREETEXT replies, ACK / REFUSE for DCL and OCL, the message status colours of the FCOM.
- FCOM entry formats and messages for every field.
- `fbw-common` MailboxBus (one line): `downlinkTransmit` on an uplink sends its existing open or failed response.
- The old placeholder ATC COM pages (connect, D-ATIS, message record) are removed.
- 33 unit tests (request frames, entry formats, ATIS text and time, message record, reports, mailbox logic).

Cockpit API Changes

New private local vars, documented in `fbw-a380x/docs/a380x-private-local-vars.md` (46 - Information Systems):

- `L:A380X_ATCCOM_ADS_STATUS` (0 ARMED, 1 CONNECTED, 2 OFF), written by the MFD, displayed by the SD mailbox.
- `L:A380X_MFD_{side}_ATCCOM_MSG_RECORD_ZOOM` (an MFD shows a recorded message: MSG RECORD USED OFFSIDE on the other one).

Screenshots (if necessary)

TO ADD: each ATC COM page next to its FCOM figure (CONNECT pages, ATIS LIST, MSG RECORD LIST and ZOOM, REQUEST with frames, REPORT, EMERGENCY); the SD mailbox with an uplink and its answer buttons.

References

- A380 FCOM DSC-46-10-20-30 (ATC COM pages), DSC-46-10-10-70 (mailbox), DSC-46-10-20-40 (ATC COM messages); the CPDLC message elements (DM / UM numbers) of the FBW ATSU.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. flyPad: set the Hoppie ACARS network and logon code. MFD ATC COM > CONNECT > NOTIFICATION: notify a center: NOTIFIED TO CENTERS, then CONNECTION STATUS.
2. ATIS: request the departure ATIS; RECEIVED shows it; AUTO UPDATE.
3. REQUEST: add CLB TO and DIRECT frames, XFR TO MAILBOX; send from the SD mailbox; the reply buttons follow the uplink.
4. MSG RECORD: the dialogue appears; ZOOM; ERASE ALL keeps the open ones.
5. EMER: MAYDAY frame, XFR: ADS EMERGENCY.
6. `npx vitest run` on the six ATC COM / mailbox spec files (33 tests).

flyPad landing calculator (A380 new, A320 reworked)

The A380 OIS LDG PERF application on the flyPad of both aircraft, with FCOM data, FMS import and BTV exit advice

Branch	pr/23-efb-landing-calculator
Aircraft	A380X and A32NX (flyPad, A380 FMC, A320 MCDU)
Commits	fc850a71e feat(efb): A380 landing calculator and reworked A320 landing calculator; e7e27cf62 fix(efb): strict null checks of the landing calculator
Base	pr/12-a320-takeoff-calculator (shared performance module and FMS links of the takeoff calculators)
Status	CI checks pass; merged into develop 224cf6741. Checked in the flyPad harness; BTV exits seen in the sim once; full in-sim test to do.

Purpose

The A380X flyPad had no landing calculator and the A32NX one had only the QRH in-flight distances, no dispatch data, no FMS import and no approach or go-around parameters. The real A380 computes landing performance in the OIS LDG PERF application (A380 FCOM PER-LND).

Specification (A380 FCOM PER-LND, DSC-32-10-30-20, LIM-32-30; A320 FCOM PER-LDG-DIS, QRH)

Part	Content
Computation	DISPATCH (factored required landing distance) or IN-FLIGHT (actual landing distance)
Runway	airport, runway, LDA, heading, elevation, slope; A380: go-around altitude
Conditions	runway condition (dry, wet, RWYCC contaminants), wind, temperature, QNH; A380: anti-ice, air conditioning
Aircraft	landing weight in t / klb with an MLW button; CONF (AUTO on the A380), go-around CONF and gradient
Landing	approach type, VLS+ increment, autoland and glide slope, braking mode (manual, autobrake, BTV on the A380), reverse thrust (greyed out without credit); A320: overweight procedure
RESULTS	LW, MLW(PERF) and its limitation (WGT / LDA / ACG), flaps, landing distance, stop margin, go-around speed and gradient, VAPP = VLS + VLS+; estimates in amber
Runway picture	touchdown, the stop point of each braking mode, the required distance; A380: BTV DRY / WET lines and the runway exits

A380 data and rules

- Landing field length chart of the Airbus A380 Aircraft Characteristics (dry, manual braking, CONF FULL, VLS), extracted from the vectors; a model fitted on it (air time 5.149 s, dry deceleration 3.253 m/s², 0.8 % RMS) for the other conditions.
- Dispatch: RLD = ALD / 0.6, wet x 1.15, contaminated = max(wet RLD, ALD x 1.15); 50 % of the headwind, 150 % of the tailwind; temperature only for water-contaminated runways or autoland; slope only for autoland (PER-LND).
- Autobrake decelerations LO / 2 / 3 / HI = 2.0 / 2.5 / 3.0 / 3.5 m/s² (DSC-32-10-30-20).
- Go-around gradient = T/W - K, calibrated on the FCOM in-flight example (386 t: 8.939 %; MLW(PERF) 569 t at the 2.7 % minimum; GA speed 141 kt).

BTV exit advice (A380, information only)

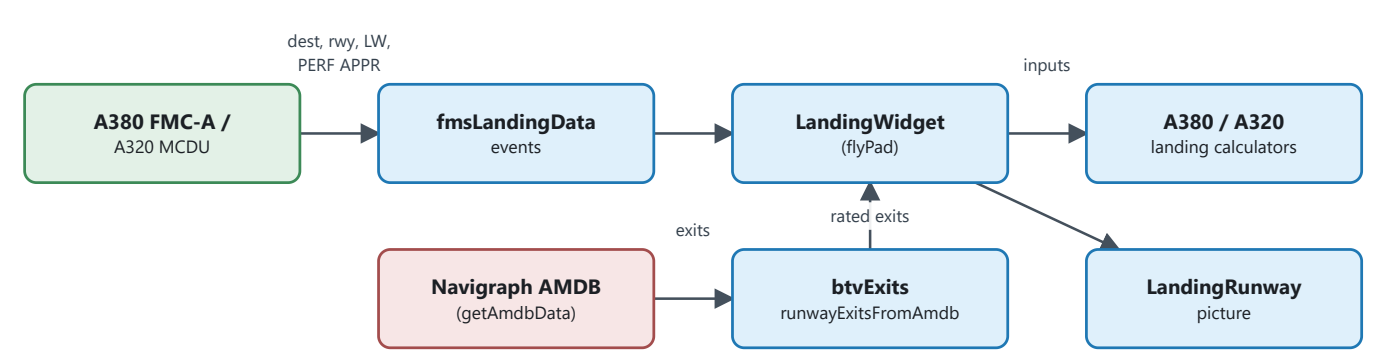
DRY / WET lines = 400 m touchdown + 5 s roll + $V^2 / (2 \times 2.8 \text{ m/s}^2 \text{ dry}, 1.8 \text{ m/s}^2 \text{ wet})$, V = TAS at VAPP minus the wind (FCOM DSC-32-10-30-20 and the FBW autobrake constants). The runway exits of the Navigraph airport map (OANS

exit rules: within 50 m of the centreline, at least 400 m after the threshold, turn up to 120 degrees) are rated: recommended (first exit beyond WET), beyond WET, dry runway only, not achievable. Nothing is shown on a contaminated runway (BTV prohibited, LIM-32-30).

A320 data

- QRH in-flight landing distances kept (x 1.15 factored shown).
- FCOM dispatch required landing distance tables (PER-LDG-DIS-RLD), autoland correction, wet minimum.
- Crosswind limits of PER-LDG-DIS-MAT; reverser credit per QRH correction.

Component added or changed by the feature MSFS / simulator Aircraft system (systems host, FMS, FWS) Display / page
Data, LVar, storage Crew, flypad, external program



Architecture

Interfaces

Name	Type	Meaning
fms_landing_data_request	event bus	the flyPad asks the FMS for its landing data (request id)
fms_landing_data	event bus	destination, runway, landing weight, QNH, temperature, wind, CONF

Tests

16 unit tests: `a380x_landing.spec.ts` (8), `a32nx_landing.spec.ts` (6), `btvExits.spec.ts` (2).

Limitations

- Only the A380 dry manual-braking chart is Airbus data; every other A380 value is a model estimate (amber).
- The BTV exits need a Navigraph airport map; the advice does not arm the BTV (that stays on the OANS).

PR description (from the PR draft)

From `pr-drafts/23-efb-landing-calculator.md`, section "Summary of Changes" onwards.

Summary of Changes

One landing screen for both aircraft, laid out after the A380 OIS LDG PERF application (A380 FCOM PER-LND): a DISPATCH / IN-FLIGHT computation type, Runway / Conditions / Aircraft / Landing sections, a RESULTS panel and a runway picture.

- **Inputs:** airport and runway (LDA, heading, elevation, slope), runway condition (dry, wet and the RWYCC contaminants), wind, temperature and QNH, anti-ice and air conditioning (A380), landing weight in tonnes (or klb) with an MLW button, CONF (AUTO on the A380), go-around CONF and gradient, approach type and VLS+ increment, autoland and glide slope, braking mode (manual, autobrake modes, BTV on the A380), reverse thrust (greyed out when the condition gives no reverser credit), overweight procedure (A320).
- **RESULTS:** landing weight, MLW(PERF) and its limitation code (weight, landing distance, go-around gradient), flaps, landing distance (factored for dispatch), stop margin, go-around speed and gradient, VAPP = VLS + VLS+; estimated values in amber.
- **Runway picture:** touchdown, the stop point of each braking mode, the required distance bar, and on the A380 the BTV DRY / WET lines with the runway exits.
- **A380 data:** the landing field length chart of the Airbus A380 AC document (dry, manual braking, CONF FULL, VLS) and a model fitted on it (0.8 % RMS) for the other conditions. FCOM rules: dispatch RLD = ALD / 0.6, wet x 1.15, contaminated = max(wet RLD, ALD x 1.15); 50 % of the headwind, 150 % of the tailwind; temperature only for water-contaminated runways or autoland, slope only for autoland; autobrake decelerations LO / 2 / 3 / HI (DSC-32-10-30-20); go-around gradient calibrated on the FCOM in-flight example.
- **BTV exit advice (A380, information only):** DRY / WET lines = 400 m + 5 s roll + the braking distance at the BTV decelerations; the runway exits of the Navigraph airport map (the OANS exit rules) rated as recommended (first exit beyond the WET line), beyond WET, dry runway only, or not achievable. Not shown on contaminated runways (BTV prohibited, LIM-32-30).
- **A320 data:** the QRH in-flight landing distances are kept; the FCOM dispatch required landing distance tables (PER-LDG-DIS-RLD), the autoland correction and the wet minimum are added; the crosswind limits of PER-LDG-DIS-MAT.
- **FMS import:** the A380 FMC-A and the A320 MCDU answer the flyPad (shared `fmsLandingData` events) with the destination, runway, predicted landing weight and the PERF APPR QNH, temperature, wind and landing configuration.
- The old `RunwayVisualizationWidget` is removed.
- 16 unit tests (A380 calculator 8, A320 calculator 6, BTV exits 2).

Cockpit API Changes

None (the FMS data travels on two new event bus topics, `fms_landing_data_request` and `fms_landing_data`).

Screenshots (if necessary)

TO ADD: the A380 IN-FLIGHT and DISPATCH screens with results, the A320 screen, the BTV lines with the exits of an airport.

References

- A380 FCOM PER-LND (landing performance, OIS LDG PERF), DSC-32-10-30-20 (autobrake and BTV), LIM-32-30; Airbus A380 Aircraft Characteristics (landing field length chart).
- A320 FCOM PER-LDG-DIS (dispatch RLD tables, autoland, crosswind), QRH in-flight landing distances.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. A380 flyPad > Performance > Landing: IN-FLIGHT, fill from the FMS (destination and runway in the F-PLN, PERF APPR filled): runway, weight and conditions are imported. CALCULATE: RESULTS and the runway picture.
2. Change the runway condition, braking mode, CONF AUTO, reverse thrust: the distances and the MLW(PERF) change; reverse thrust is greyed out when it gives no credit.
3. With a Navigraph airport map: the BTV lines and the exits, with their rating; a contaminated runway hides them.
4. DISPATCH: the factored required landing distance.
5. A320 flyPad: the same screen with the A320 data; DISPATCH uses the FCOM tables.
6. `npx vitest run` on `a380x_landing.spec.ts`, `a32nx_landing.spec.ts`, `btvExits.spec.ts` (16 tests).

flyPad descent calculator

The A380 OIS IN-FLT PERF DES module on the flyPad of both aircraft, computed with the FMS performance model

Branch	pr/24-efb-descent-calculator
Aircraft	A380X and A32NX (flyPad, A380 FMC, A320 MCDU)
Commits	469b9a0c8 feat(efb): descent calculator in place of the top of descent calculator
Base	pr/23-efb-landing-calculator (FMS data answers, performance store and texts)
Status	CI checks pass; merged into develop 99f00b531. Checked in the flyPad harness; in-sim test to do.

Purpose

The flyPad Top of Descent tab only applied a descent angle to a ground speed. The real A380 computes descents in the OIS IN-FLT PERF DES module (A380 FCOM PER-IFT-DES), with the aircraft performance, speeds and conditions.

Specification (A380 FCOM PER-IFT-DES, PER-IFT-DES-STD; A320 FCOM PER-DES-STD)

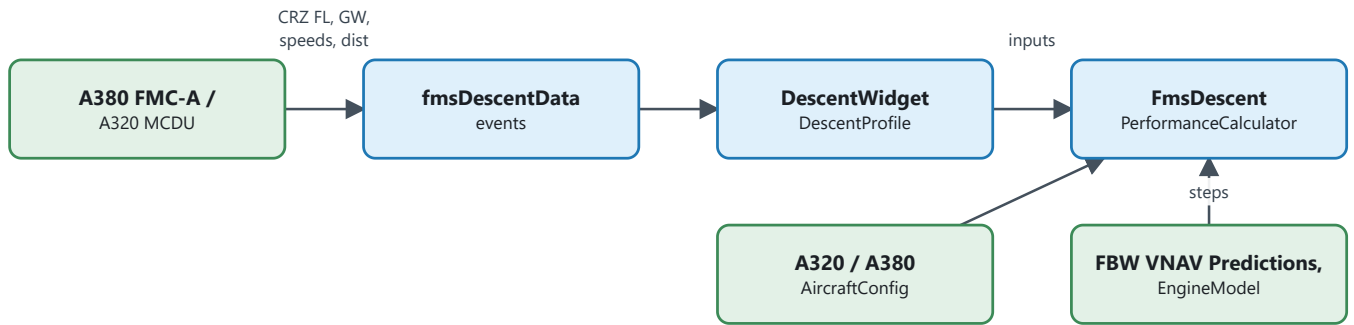
Part	Content
Descent type	ECON (FMS managed Mach / CAS), STANDARD (A380 M.85 / 300 / 250 kt below FL100, A320 M.78 / 300 / 250), GIVEN V/S, EMERGENCY (MMO / VMO with the speed brakes)
Descent	initial altitude (A/C), target altitude, speed brakes
Speed schedule	MACH, SPD, SPD LIM, LIM ALT; empty = the standard schedule
Conditions	gross weight, ISA deviation (A/C from the SAT), wind component, anti-ice (off / engine / total), fuel factor
FMS import	cruise level, destination elevation + 1500 ft, gross weight, speed limit, ECON speeds, distance to the destination
RESULTS	time, distance, fuel, mean V/S and FPA, 3 x altitude rule; crossover and deceleration
Profile	chart (T/D, CROSSOVER, DECEL, SPD LIM, the aircraft) or table every 5000 ft (time, distance, fuel, CAS, Mach, TAS, V/S, FPA)
DESCENT CHECK	distance to the target vs the profile: start in X NM (Y min), on profile, or late with the V/S and FPA needed

Model

- The FBW VNAV prediction steps (`Predictions.altitudeStep`, `verticalSpeedStep`, `speedChangeStep`, idle N1 of the engine model) with the A320 / A380 aircraft config, integrated in 1000 ft steps, so the flyPad agrees with the FMS descent predictions.
- Deceleration to the speed limit on the -1 degree idle path of the FMS descent strategy, ending at the limit altitude.
- GIVEN V/S steeper than the idle descent: idle there, with a warning (the pressure altitude below which it applies).
- Anti-ice: the corrections of the A320 FCOM descent table, marked as estimates (amber).
- Check (A320, M.78 / 300 / 250, 65 t, FL390 to 1500 ft): FCOM 106 NM / 17.4 min / 165 kg, calculator 114 NM / 18.8 min / 146 kg (the FCOM table is for the CFM56 A320, the model is the A320neo).

 Component added or changed by the feature  MSFS / simulator  Aircraft system (systems host, FMS, FWS)  Display / page

 Data, LVar, storage  Crew, flypad, external program



Architecture

Interfaces

Name	Type	Meaning
fms_descent_data_request	event bus	the flyPad asks the FMS for its descent data (request id)
fms_descent_data	event bus	cruise level, destination and elevation, distance, gross weight, cost index, ECON speeds, speed limit

Tests

3 unit tests in `FmsDescentPerformanceCalculator.spec.ts`: close to the A320 FCOM table; wind, temperature and anti-ice; EMERGENCY and the given V/S check.

Limitations

- The deceleration to 250 kt takes about 5000 ft, as in the FBW FMS model (a -1 degree idle path).
- No bleed in the engine model: the anti-ice effect comes from the A320 FCOM table on both aircraft.
- The old TODCalculator components are removed; its store slice is left in place, unused.

PR description (from the PR draft)

From `pr-drafts/24-efb-descent-calculator.md`, section "Summary of Changes" onwards.

Summary of Changes

The Top of Descent tab of the flyPad performance page becomes a descent calculator, after the A380 OIS IN-FLT PERF DES module (A380 FCOM PER-IFT-DES), on both aircraft.

- **Descent types:** ECON (the managed descent Mach and CAS of the FMS), STANDARD (the FCOM speed schedule: A380 M.85 / 300 kt / 250 kt below FL100, PER-IFT-DES-STD; A320 M.78 / 300 kt / 250 kt, PER-DES-STD), GIVEN V/S, EMERGENCY (MMO / VMO with the speed brakes).
- **Inputs:** initial altitude (A/C: the current one) and target altitude; the speed schedule (MACH, SPD, SPD LIM, LIM ALT; empty = the standard one); gross weight, ISA deviation (A/C: from the current SAT), wind component, anti-ice (off / engine / total), fuel factor, speed brakes. **Fill data from FMS:** cruise level (or the current altitude when below), destination elevation + 1500 ft, gross weight, the FMS speed limit, the ECON speeds and the distance to the destination.
- **RESULTS:** time, distance, fuel, mean V/S and flight path angle, and the 3 x altitude rule for comparison; the crossover altitude and the deceleration to the speed limit.
- **Profile:** a chart of altitude against the distance to the target (T/D, crossover, DECEL, SPD LIM) or a table every 5000 ft (time, distance, fuel, CAS, Mach, TAS, V/S, FPA), like the FCOM results table.
- **DESCENT CHECK:** with the distance to the target, the aircraft altitude and ground speed: start the descent in X NM (Y min), on profile, or N NM late with the V/S and FPA needed; the aircraft is drawn on the profile.
- **Model:** the FBW VNAV prediction steps (`Predictions.altitudeStep`, `verticalSpeedStep`, `speedChangeStep`, with the idle N1 of the engine model) and the aircraft config of each aircraft, integrated in 1000 ft steps; the deceleration to the speed limit on the -1 degree idle path of the FMS descent strategy, so the flyPad agrees with the FMS predictions. GIVEN V/S steeper than idle: idle there, with a warning. Anti-ice: the corrections of the A320 FCOM descent table, shown as estimates (amber).
- **FMS import:** the A380 FMC-A and the A320 MCDU answer the flyPad (shared `fmsDescentData` events).
- The old `TODCalculator` components are removed (its `todCalculator` store slice is left in place; nothing reads it any more).
- 3 unit tests (close to the A320 FCOM descent table; wind, temperature and anti-ice; EMERGENCY and the given V/S check).

Check against the A320 FCOM descent table (M.78 / 300 / 250, 65 t, FL390 to 1500 ft): FCOM 106 NM / 17.4 min / 165 kg, calculator 114 NM / 18.8 min / 146 kg. The FCOM table is for the CFM56 A320; the FBW A320neo model descends about 8 % longer.

Cockpit API Changes

None (two new event bus topics, `fms_descent_data_request` and `fms_descent_data`).

Screenshots (if necessary)

TO ADD: the A380 screen with the profile chart and the table, the A320 screen, the DESCENT CHECK late and early.

References

- A380 FCOM PER-IFT-DES (IN-FLT PERF DES module, descent types, speed profile, conditions), PER-IFT-DES-STD.
- A320 FCOM PER-DES-STD (standard descent table and its anti-ice corrections).

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. flyPad > Performance > Top of Descent, in cruise with a flight plan: Fill data from FMS: the cruise level, target, weight, speed limit and distance are filled. CALCULATE: RESULTS and the profile.
2. ECON: the speeds of the FMS; compare the T/D with the one of the ND.
3. GIVEN V/S 3000 ft/min, EMERGENCY: shorter descents; a V/S steeper than idle gives the warning.
4. DESCENT CHECK: start in X NM before the T/D; late after it, with the V/S and FPA needed.
5. `npx vitest run fbw-a32nx/src/systems/fmgc/src/performance/FmsDescentPerformanceCalculator.spec.ts` (3 tests).